

**MAHATMA GANDHI UNIVERSITY
UNDER GRADUATE VOCATIONAL
PROGRAMMES (HONOURS)
SYLLABUS
MGU-B.VOC.(Honours)**

(2025 Admission Onwards)



Faculty: Technology and Applied Sciences

Expert Committee: Emerging Technology and Digital Proficiency

Programme: B.Voc. (Honours) Industrial Instrumentation and Automation

**Mahatma Gandhi University
Priyadarshini Hills
Kottayam – 686560,
Kerala, India**

CONTENTS

Sl.No	TITLE	Page No
1	Preface	3
2	Expert Committee & Syllabus workshop participants	4
3	Syllabus Index	6
4	Job Roles & Qualification Pack for Certificate, Diploma, Bachelor's and Honours Degrees.	10
5	Semester 1	11
6	Semester 2	23
7	Semester 3	45
8	Semester 4	70
9	Semester 5	104
10	Semester 6	145
11	Semester 7 & 8	179
12	Apprenticeship	181
13	Research internship	184
14	On-the-Job training	187

Preface

The primary objective of **B.Voc.(Honours) Industrial Instrumentation and Automation** is to equip students with the theoretical knowledge, practical expertise, and professional acumen necessary to thrive in the domains of industrial instrumentation, process automation, control systems, and related technologies. Through an outcome-based approach, the syllabus emphasizes analytical thinking, technical mastery, adaptability, and innovative problem-solving—key attributes essential for navigating the challenges and opportunities of Industry 4.0 and beyond.

Industry-Aligned Curriculum: The syllabus is designed to reflect contemporary trends, tools, and technologies in industrial automation and instrumentation, preparing graduates for careers in manufacturing, process industries, automation design, maintenance, and system integration.

Focus on Core Engineering Concepts: Core subjects such as sensors and transducers, industrial control systems, programmable logic controllers (PLCs), SCADA, and instrumentation practices form the backbone of the curriculum, providing a strong technical foundation.

Hands-on and Applied Learning: The program emphasizes experiential learning through laboratory sessions, simulation-based training, field visits, live projects, and industrial internships—ensuring that students develop real-world competencies and job-readiness.

Interdisciplinary and Practical Integration: The syllabus blends essential knowledge from electronics, control engineering, data acquisition, communication protocols, and safety standards, providing a holistic understanding of industrial systems.

Continuous Assessment and Feedback: The curriculum integrates continuous evaluation, project assessments, and structured feedback mechanisms to monitor student progress and foster self-directed learning and performance improvement.

Adaptability and Future-Readiness: Designed with flexibility in mind, the syllabus allows for timely updates in response to technological advancements, shifting industry demands, and stakeholder feedback—ensuring sustained relevance and applicability.

Expert Committee and Syllabus Workshop Participants

Expert Committee – Emerging Technology and Digital Proficiency			
Sl:No	Name of Participant	Designation	Address
1	Dr. Leena C.Sekhar(Convener)	Associate Professor of Computer Applications & HOD	MES College, Nedumkandam
2	Dr. Shereena V. B.	Associate Professor of Computer Science	MES College, Nedumkandam
3	Dr. Antu Annam Thomas	Assistant Professor of Computer Applications & HOD	Mar Thoma College, Thiruvalla
4	Dr. Jina Varghese	Assistant Professor of Computer Applications	Baselios Paulose II Catholicose College,Piravom
5	Meera Nair	Assistant Professor of Computer Applications	NSS College, Rajakumari
6	Soumya M V	Assistant Professor of Computer Science	SAS SNDP Yogam College, Konni
7	Resmi Cherian	B.Voc Nodal Officer	CMS College, Kottayam
8	Christina Merin Babu	Assistant Professor (B.Voc)	Mar Athanasius College, Kothamangalam
9	Amruthavarshini R	Assistant Professor (B.Voc)	St. Xaviers College for Women, Aluva
10	Raji S. Pillai	B.Voc Nodal Officer	St. Theresa's College, Ernakulam
11	Spasiba Raveendran	Assistant Professor of Computer Science	SAS SNDP Yogam College, Konni

12	Syam S.	Assistant Professor of Computer Applications	Sree Narayana Arts & Science College, Kumarakom, Kottayam
13	Dr. Abahayadev M.	Assistant Professor of Computer Science	MES College, Nedumkandam
14	Shanavas C.T.	Assistant Professor of Computer Science	MES College, Marampilly
Syllabus Workshop Participants			
15	Smitha Salim	Assistant Professor of B.Voc Industrial Instrumentation and Automation & HOD	MES College Marampally Email: nmsmitha@gmail.com Contact: +919895097461
16	Smitha K	Assistant Professor of B.Voc Industrial Instrumentation and Automation	MES College Marampally Email: smithaprasadk191275@gmail.com Contact: +917561865217
17	Shalima Kareem	Assistant Professor of B.Voc Industrial Instrumentation and Automation	MES College Marampally Email: kareemshalima@gmail.com Contact: +919400870928



MGU-B.VOC. (HONOURS)

Syllabus

Syllabus Index

Name of the Major Subject: Industrial Instrumentation and Automation

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG1SDCIIM100	Hybrid Electronics	SDC	4	5	3	2	0
MG1SDCIIM101	Automation Fundamentals	SDC	4	5	3	2	0
MG1SDCIIM102	Mathematics for Instrumentation	SDC	4	4	4	0	0
MG1MDCIIM100	Robotics	MDC	3	3	3	0	0
MG1SDCIIM103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On -the -Job Training

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG2SDCIIM100	Biomedical Instrumentation	SDC	4	4	4	0	0
MG2SDCIIM101	Electronic circuits Design	SDC	4	5	3	2	0
MG2SDCIIM102	Smart Sensors	SDC	4	5	3	2	0
MG2MDCIIM100	3- Dimensional Printing Technology	MDC	3	4	2	2	0
MG2SDCIIM103	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On -the -Job Training

Semester: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/week	Hour Distribution /week		
					L	P	O
MG3SDCIIM200	Industrial Instruments	SDC	4	4	4	0	0
MG3SDCIIM201	Printed Circuit Board Designing and Simulation	SDC	4	5	3	2	0
MG3SDCIIM202	Programmable Logic Devices	SDC	4	5	3	2	0
MG3MPCIIM200	Measurement Technology	MPC	4	5	3	2	0
MG3MDCIIM200	E-Waste Management systems in Kerala	MDC	3	3	3	0	0
MG3SDCIIM203	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On -the- Job Training

Semester: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours/Week	Hour Distribution /week		
					L	P	O
MG4SDCIIM200	Fibre Optic Technology	SDC	4	4	4	0	0
MG4SDCIIM201	Final Control Elements	SDC	4	5	3	2	0
MG4SDCIIM202	Embedded Automation	SDC	4	5	3	2	0
MG4MPCIIM200	Industrial Robotics	MPC	4	5	3	2	0
MG4SECIIM200	Information Technology (IT) for entrepreneurs	SEC	3	3	3	0	0
MG4VACIIM200	Environmental studies for Instrumentation	VAC	3	3	3	0	0
MG4INTIIM200	Internship	INT	2	60 hours			

L — Lecture, P — Practical/Practicum , O — On -the -Job Training

Semester: 5

Course Code	Title of the Course	Type of the Course	Credit	Hours/week	Hour Distribution /week		
					L	P	O
MG5SDCIIM300	Industrial Communication Protocols	SDC	4	4	4	0	0
MG5SDCIIM301	AI based Robotic Automation	SDC	4	5	3	2	0
MG5SDEIIM300	Process Control Instrumentation	SDE (Any One)	4	5	3	2	0
MG5SDEIIM301	Industry Layouts						
MG5SDEIIM302	Power Engineering						
MG5SECIIM300	Virtual Instrumentation	SEC	3	4	2	2	0
MG5MPCIIM300	Process Control Communication	MPC	4	4	4	0	0
MG5VACIIM300	Digital Empowerment	VAC	3	3	3	0	0

L — Lecture, P — Practical/Practicum , O — On- the- Job Training

Semester: 6

Course Code	Title of the Course	Type of the Course	Credit	Hours/week	Hour Distribution /week		
					L	P	O
MG6SDCIIM300	Applications of Programmable Logic Controller (PLC) and Supervisory Control and Data Acquisition (SCADA) in Industry	SDC	4	5	3	2	0
MG6SDEIIM300	Advanced RISC Machine (ARM) and VEGA Processors	SDE (Any One)	4	5	3	2	0
MG6SDEIIM301	Communication Electronics						
MG6SECIIM300	Solar Technology	SEC	3	3	3	0	0

MG6MPCIIIM300	Automation Controllers	MPC	4	4	4	0	0
MG6VACIIIM300	Industrial Safety Measures	VAC	3	3	3	0	0
MG6PRJIIIM300	Project	PRJ	4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On -the -Job Training

Semesters: 7 & 8

B.Voc Honours							
Course Code	Title of the Course	Type of the Course	Credits	Number Of Days	Credit Distribution		
					L	P	O
MG7APPIIM400	Apprenticeship	APP	28	280	0	28	0
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On- the -Job Training

B.Voc Honours with Research							
Course Code	Title of the Course	Type of the Course	Credits	Number Of Days	Credit Distribution		
					L	P	O
MG7RINIIM400	Research Internship	RIN	20	200	0	20	0
NA	Online	SDC	4	NA			
NA	Online	SDC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On -the -Job Training

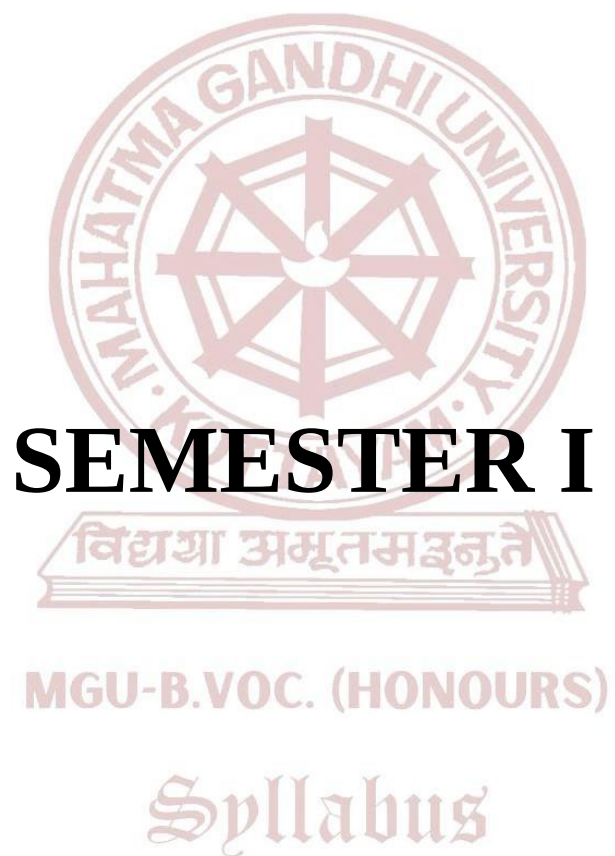
Job Roles and Qualification Packs for Certificate, Diploma, Bachelor's, and Honours Degrees.

Job Roles	NHEQF Level	QPs Aligned	Sector Skill	Year
Industrial Automation Technician	Level 4.5	IAS/Q5601	IASC SSC	First Year
Industrial Automation Specialist	Level 5.0	IAS/Q8005	IASC SSC	Second Year
Industrial Automation System Designer	Level 5.5	IAS/Q8006	IASC SSC	Third Year
Instrumentation and Control Engineer	Level 6.0	ELE/Q6201	ELE SSC	Fourth Year



MGU-B.VOC. (HONOURS)

Syllabus



semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG1SDCIIM100	Hybrid electronics	SDC	4	5	3	2	0
MG1SDCIIM101	Automation Fundamentals	SDC	4	5	3	2	0
MG1SDCIIM102	Mathematics for Instrumentation	SDC	4	4	4	0	0
MG1MDCIIM100	Robotics	MDC	3	3	3	0	0
MG1SDCIIM103	OJT	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc. (Honours) Industrial Instrumentation and Automation				
Course Name	Hybrid Electronics				
Type of Course	SDC				
Course Code	MG1SDCIIM100				
Course Level	100				
Course Summary	Helps students understand basic electronics, semiconductor devices, and digital systems, and develop skills to identify components and design simple electronic circuits.				
Semester	1	Credits			Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand and explain the fundamental principles of electronics and the structure and behavior of semiconductor materials and devices..	U	1,2,3,4,6,7,8,9,10
2	Understand the working principles, characteristics, and applications of diodes and transistors.	U	1,2,3,4,6,7,8,9,10
3	Understand and apply number systems, Boolean algebra, combinational and sequential logic circuits for the design and analysis of basic digital systems..	A	1,2,3,4,6,7,8,9,10
4	Able to identify electronic components, analyze diode and transistor characteristics, and design basic digital logic circuits.	An	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	3	3	2	1	0	1	1	1	1	2
CO 3	3	3	2	1	0	1	1	1	1	3
CO 4	3	3	2	2	1	1	1	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of Electronics and Semiconductor Devices			
	1.1	Basic Electronic Components: Passive Components: Active Components	2	1
	1.2	Basics of Electricity: Electric field, charge, current, and potential, Resistance and Ohm's Law.	4	1

	1.3	Structure of Solids and Semiconductors: : Conductors, Insulators, Semiconductors. Energy bands: valence and conduction bands	4	1
	1.4	Intrinsic vs. Extrinsic semiconductors: N-type and P-type semiconductors, mobility, conductivity, energy gap, drift and diffusion current	5	1
	Semiconductor Devices and Applications			
	2.1	Diodes and Their Applications, PN Junction Diode: structure, working, VI characteristics.	5	2
	2.2	Zener Diode: Reverse bias characteristics, Applications in voltage regulation	3	2
2	2.3	Transistors Bipolar Junction Transistor (BJT):Types: NPN & PNP- Construction, operation, and biasing.	3	2
	2.4	Configurations: CB, CE, CC – input/output characteristics Terminal identification and testing	4	2
	Basic Concepts of Digital Electronics			
	3.1	Number Systems: Decimal, Binary, Octal, Hexadecimal – conversions between systems	2	3
3	3.2	Boolean Algebra and Logic Gates, Binary Logic Functions: AND, OR, NOT, NAND, NOR, XOR, XNOR Boolean Laws and Theorems: Truth Tables and Logic Simplification DeMorgan's Theorem.	5	3

	3.3	Combinational Logic Circuits, Encoders and Decoders, Multiplexers (MUX) and Demultiplexers (DEMUX) Adders: Half Adder, Full Adder	4	3
	3.4	Sequential Logic Circuits: Flip-Flops, Counters, Registers.	4	3
4	(Practical Session)			
	Lab Experiments			
	4.1	1. To identify and test passive and active components and measure their values. 2. To verify Ohm's Law and calculate the resistance of a given resistor 3. To study the forward and reverse bias characteristics of a PN junction diode. 4. Identify terminals and test the working condition of diodes and BJTs using a multimeter.	10	4
	4.2	5. To study the input and output characteristics of a BJT in CE configuration. 6. To study the reverse bias characteristics of a Zener diode and its application in voltage regulation. 7. Familiarization of logic gates AND, OR, NOT, NAND, NOR, XOR, XNOR.	20	4

		8. Realization of Boolean Expressions using Gates. 9. Implementation of Half Adder and Full Adder. 10. Implement and verify multiplexer/demultiplexer logic.		
5	Teacher specific content			

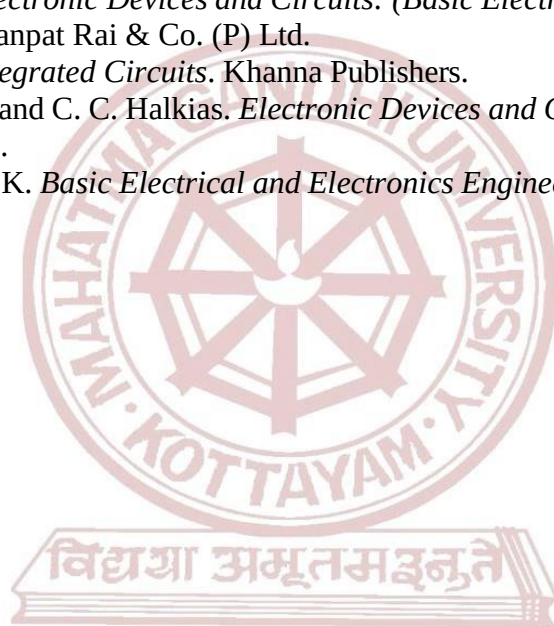
Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES:

1. Theraja, B. L., and A. K. Theraja. *Textbook of Electrical Technology: Basic Electrical Engineering in S. I. Units*. Vol. 1, 1st rev. ed., Schan.
2. Bhargava, N. N., D. C. Kulshreshtha, and S. C. Gupta. *Basic Electronics and Linear Circuits*. 2nd ed., Kindle ed., Tata McGraw-Hill Education.
3. Maini, Anil K. *Digital Electronics: Principles and Integrated Circuits*.

SUGGESTED READINGS:

1. Sahdev, S. K. *Electronic Devices and Circuits: (Basic Electronics; EDC; Power Electronics)*. Dhanpat Rai & Co. (P) Ltd.
2. Botkar, K. R. *Integrated Circuits*. Khanna Publishers.
3. Millman, Jacob, and C. C. Halkias. *Electronic Devices and Circuits*. Tata McGraw-Hill Publications.
4. Bhattacharya, S. K. *Basic Electrical and Electronics Engineering*. Kindle ed., Pearson Education.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Automation fundamentals				
Type of Course	SDC				
Course Code	MG1SDCIIM101				
Course Level	100				
Course Summary	Provides foundational knowledge and practical skills in the measurement and control of physical variables in industrial and laboratory settings. This course introduces the principles, devices, and techniques used to measure parameters such as temperature, pressure, flow, level, and electrical quantities.				
Semester	1	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre-requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify different types of instrumentation systems.	U	1,2,3,4,5,9,10
2	Understand characteristics of Measurement System	A	1,2,3,4,8,9,10
3	Understand the working of analog and digital Instruments	E	1,2,3,4,8,9,10
4	Familiarize with the hand tools used in the Industry	S	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	2	1	1	0	0	0	2	1
CO 2	3	3	2	1	0	0	0	1	1	2
CO 3	3	3	2	1	0	0	0	1	2	2
CO 4	2	2	3	2	1	1	1	1	2	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of measurement system			1
	1.1	Fundamentals of measurement system, functions of instruments and measuring systems	5	
	1.2	Classification of instruments ,mechanical, electrical and electronic instruments	5	
	1.3	Applications of measurement system, types of instrumentation systems.	5	
2	Characteristics of Measurement System			2
	2.1	Characteristics of Measurement System. Static and Dynamic characteristics of Measurement system	5	
	2.2	Scale range scale span reproducibility, drift, repeatability, noise, sources of noise, accuracy, precision, speed of response, measuring lag, fidelity	5	
	2.3	Errors Types and sources of errors, error reduction techniques, dynamic error.	5	
3	Analog and Digital Instruments,			3
	3.1	Galvanometer (D'Arsonval), analog Ammeters & Voltmeters, and, watt meters, energy meter (principle, working and construction)	5	
	3.2	Bridge circuits- D.C. bridges – Wheatstone's bridge, Kelvin bridge, A.C. bridges- Maxwell bridge	5	

	3.3	Digital Tachometer, Ultrasonic Distance meter, Digital Thermometer DSO, Frequency meter. Digital vs Analog instruments Digital volt meter	5	3
4	Familiarization with tools handheld tools			
	4.1	1.Familiarize handheld tools with neat figures 2.Measure diameter Meter of given work piece using Vernier 3. Measure diameter Meter of given work piece using Screw gauge	10	4
	4.2	4.Drill a hole on the given wooden piece and measure depth & diameter Meter 5. Use hand drill to make a groove and measure depth 6.Speed measurement by using Tachometer	10	
	4.3	7.Distance measurement using ultrasonic meter 8.Angle measurement using clinometers 9.Familiarize CRO	10	
5	Teacher Specific Content			

Syllabus

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT Mode of Assessment
Assessment Types	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1.Lab involvement 2.Record 3.Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

Syllabus

REFERENCES

1. Albert D. Helfrick *Modern electronics Instrumentation and measurement techniques*
2. Umesh Sinha, Satyaprakashan, *Electrical and Electronics measurements and instrumentation*, Tech India publication 1992

SUGGESTED READINGS

1. R.K. Rajput, *Basic mechanical engineering*, Laxmi Publications
2. R.S.Khurmi,J.K.Gupta, *Work shop technology*, S.Chand publishers
3. R.Raman, *Elements of precision engineering*, Oxford & IBH Publishing New Delhi



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Mathematics for Instrumentation				
Type of Course	SDC				
Course Code	MG1SDCIIM102				
Course Level	100				
Course Summary	This course gives an introduction to basic concepts of vector algebra. Students will get a good understanding of matrix operations and basic concepts of determinants. This course offers an introductory session to calculus. Students acquire knowledge to construct Boolean functions which can be applied in various logical circuits.				
Semester	1	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	60
Pre-requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Apply the notions of Vector for solving problems in different fields of science.	A	1,2,3,10
2	Apply the concept of matrix for solving linear equations	A	1,2,3,10
3	Analyze the theory of Calculus with real life problems	An	1,2,3,6,10
4	Analyze the truth tables and logical expressions associated with each logic gates	An	1,2,3,6,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	0	0	0	0	0	0	2
CO 2	3	3	1	0	0	0	0	0	0	2
CO 3	3	3	2	0	0	1	0	0	0	2
CO 4	3	3	2	0	0	1	0	0	1	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Vector Algebra			
	1.1	Vectors – Definition, Addition, Subtraction, Multiplication	5	1
	1.2	Dot Product – Definition, Geometrical interpretations, Simple problems for finding Dot products	5	1
	1.3	Cross Product – Definition, Geometrical interpretations, Simple problems for finding Cross products	5	1
Text 2: Chapter 3 section 1, 5, 6 (Relevant sections only. Proofs of all theorems excluded)				
2	Matrices and Determinants			
	2.1	Definition and different types of Matrices	5	2
	2.2	Matrix operations and related Matrices	5	2
	2.3	Definition of Determinants. Solution of consistent non-homogeneous linear equations using Cramer's rule and Matrix inversion methods (only of order 2x2)	5	2
Text 2: Chapter 2 sections 2,4,5,6,9 (Relevant sections only, proof of all theorems excluded, problems of inverse must be of order 2x2)				
3	Calculus			
	3.1	Introduction to techniques of Differentiation	5	3
	3.2	Product and Quotient rules	5	3

3	3.3	Antiderivatives, Indefinite integrals and their properties	5	3
Text 3: Chapter 2 sections 2,3 Chapter 4 section 2 (Relevant sections only, proof of all theorems excluded, simple problems from polynomial functions only)				
4	Boolean Algebra			
	4.1	Boolean Function	5	4
	4.2	SOP form of Boolean expression	5	4
	4.3	Logic Gates and representations	5	4
Text 1: Chapter 10 sections 1,2,3 (Relevant sections only, proof of all theorems excluded, problems of inverse must be of order 2x2)				
5	Teacher Specific Contents			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Brainstorming Lecture, Explicit Teaching, Active Cooperative Learning							
Assessment Types	MODE OF ASSESSMENT							
	Mode of Assessment							
	A. Continuous Comprehensive Assessment (CCA): 30 marks							
	<table> <tr> <th>Components</th><th>Mark Distribution</th></tr> <tr> <td>Test</td><td>15 marks</td></tr> <tr> <td>Assignment</td><td>10 marks</td></tr> <tr> <td>Viva voce</td><td>5 marks</td></tr> </table>	Components	Mark Distribution	Test	15 marks	Assignment	10 marks	Viva voce
Components	Mark Distribution							
Test	15 marks							
Assignment	10 marks							
Viva voce	5 marks							
	B. End Semester Evaluation (ESE) ESE for Theory : Written Test(70 Marks , 2 hrs) Part A: Very Short Answer Questions (5 out of 8 Questions)- (5*2=10 Marks) Part B: Short Answer Questions (5 out of 8 Questions) - (5*6=30 Marks) Part C:Essay Questions(2 out of 4 Questions) - (2*15=30 Marks)							

REFERENCES

1. Rosen, Kenneth. H. Discrete Mathematics and its applications, 6th edition, McGraw Hill Publishing Co. New Delhi, 2006.
2. Grewal, B. S. Higher engineering Mathematics, 42th Edition, Khanna publications, 2012.
3. Anton, Howard, Irl Bivens, Stephen Davis. Calculus. 10th ed. John Wiley & Sons, Inc., 2012.

SUGGESTED READINGS

1. Kreyszig, Erwin. Advanced Engineering Mathematics, Wiley, India, 2006.
2. Thomas, George B., Jr., and Maurice D. Weir. Thomas' Calculus. 12th ed. Pearson, 2009.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Robotics			
Type of Course	MDC			
Course Code	MG1MDCIIM100			
Course Level	100			
Course Summary	A hands-on introduction to Arduino-based robotics covering design, sensors, and control, aimed at enabling learners to build practical robotic solutions and develop skills to instruct and guide others in robotics applications.			
Semester	1	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		3	0	45
Pre- requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the fundamentals of Arduino-based robotics ,basic robot assembly and programming for functional robotic applications..	U	1,2,3, 4,5,6, 7,8,9, 10
2	Understand and explain the fundamental concepts of robotics, including sensors, actuators, and microcontroller integration.	A	1,2,3,4,5,6, 7,8,9,10

3	Design and assemble simple robotic systems using Arduino and basic electronic components.	U	1,2,3,4,5,6,7,8,9,10
4	Develop and implement basic control algorithms to operate and troubleshoot robotic systems for real-world tasks	U	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	3	1	1	1	2	1	2	3
CO 2	3	3	3	1	1	1	1	1	2	3
CO 3	3	3	3	2	2	1	1	1	3	3
CO 4	3	2	2	3	3	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Robotics			
	1.1	Basics of Electronics and Arduino, Introduction to electricity, voltage, current, and resistance	5	1
	1.2	Overview of Arduino board and IDE	6	1
	1.3	Interfacing LEDs, buzzers, switches, and basic sensors, Writing and uploading simple Arduino programs.	4	1

2	2.1	Sensors and Actuators Working principles of common sensors (IR, ultrasonic, temperature, etc.) Introduction to actuators: motors (DC, servo), relays. Sensor integration with Arduino Reading sensor data and controlling actuators	2	2
	2.2	Obstacle avoiding robot, Basics of motor driver circuits (e.g., L298N)	5	2
	2.3	Programming logic and troubleshooting techniques Building and Programming Simple Robots Line follower robot Obstacle avoiding robot	8	2
	Practical Session			
	Lab experiments			
3	3.1	1. Blinking an LED using Arduino 2. Interfacing Push Button with Arduino 3. Controlling a Buzzer and LED using a Switch 4. Reading Analog Sensor (e.g., Potentiometer or LDR) 5. Ultrasonic Sensor for Distance measurement.	8	3
	3.2	6. Motor Control using L298N Motor Driver 7. Line Follower Robot using IR Sensors 8. Obstacle Avoiding Robot using Ultrasonic Sensor 9. Servo Motor Control 10. Bluetooth-Controlled Robot using HC-05 Module	7	3
4	Teacher Specific Contents			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical(25 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination (50 Marks, 1.5 hrs) Part A : Short Answer Questions(5 out of 7 Questions) - (5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20Marks) Record : 10 Marks

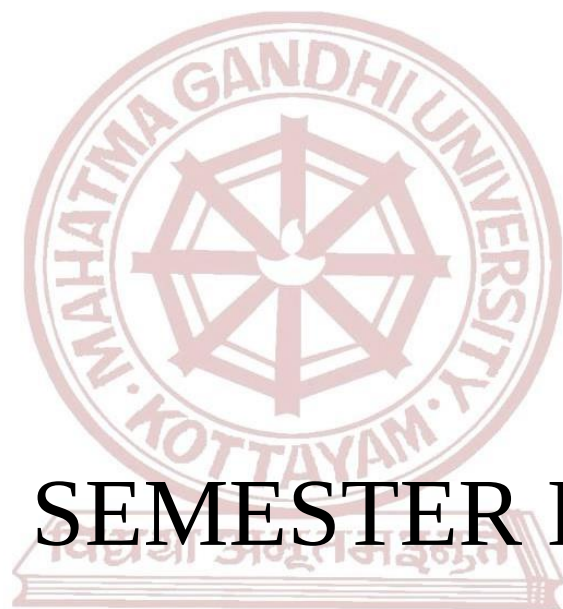
Syllabus

REFERENCES

1. Kenneth J Waldron, *Kinematics, Dynamics & Design of Machinery*, Gary L. Kinzel, John Wiley & Sons, India.
2. Craig, *Introduction to Robotics*, Pearson Prentice Hall

SUGGESTED READINGS:

1. Murray, R.M., Li, Z., and Sastry, S.S., *A Mathematical Introduction to Robotic Manipulator*, CRC Press, 1994.
2. Jeremy Blum, *Exploring Arduino: Tools and Techniques for Engineering Wizardry* Paperback – 9 Aug 2013
3. Erik Savasgard, *Arduino: 101 Beginners Guide* Paperback – July 29, 2015
4. John-David Warren, Josh Adams, Harald Molle, *Arduino Robotics* 1st ed. Edition



SEMESTER II

MGU-B.VOC. (HONOURS)

Syllabus

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/week	Hour Distribution /week		
					L	P	O
MG2SDCIIM100	Biomedical Instrumentation	SDC	4	4	4	0	0
MG2SDCIIM101	Electronic circuits Design	SDC	4	5	3	2	0
MG2SDCIIM102	Smart Sensors	SDC	4	5	3	2	0
MG2MDCIIM100	3-Dimensional Printing Technology	MDC	3	4	2	2	0
MG2SDCIIM103	On-the-Job Training	SDC	2	5	0	0	5

Syllabus

L — Lecture, P — Practical/Practicum , O — On the Job Training



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Biomedical Instrumentation			
Type of Course	SDC			
Course Code	MG2SDCIIM100			
Course Level	100			
Course Summary	Biomedical Instrumentation is an interdisciplinary course that combines engineering principles with biological and medical sciences to develop diagnostic and therapeutic devices. The course covers the principles, design, operation, and application of instruments used in medical settings to monitor, diagnose, and treat patients.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		4	0	
			OJT	
			0	60
Pre-requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basic principles of different instruments/equipment used in the health care industry	U	1,2,3,4,10
2	Analyze the electrical activity of the heart and its representation through an ECG.	A	1,2,3,4,10
3	Understand the principles of Doppler and color flow imaging, and their applications in vascular and cardiac imaging	U	1,2,3,4,10
4	Understand and explain the methods for blood pressure measurement	U	1,2,3,4,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	1	1	0	0	0	0	0	2
CO 2	3	3	1	1	0	0	0	0	0	2
CO 3	3	3	2	1	0	0	0	0	0	2
CO 4	2	2	1	2	0	0	0	0	0	2

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Bio potential			
	1.1	Resting and action potentials - propagation of action potentials – Examples of bioelectric potentials.	5	1
	1.2	ECG, EEG, EMG, Theory	5	
	1.3	Electrodes for measurement of biopotential Transducers for measurement of temperature, pressure & displacement -	5	
2	Electrical activity of Heart			
	2.1	Electrocardiogram - lead systems - ECG machine –	5	2
	2.2	Block diagram Cardiac pacemakers – internal and external pacemakers	5	
	2.3	Defibrillators – basic principles. Measurement of heart sounds – phonocardiography	5	
	Ultrasonic imaging			

3	3.1	Basic principles - Ultrasonic transducers & Types - modes of image display	5	3
	3.2	--Principles & applications. Doppler & colour flow imaging MRI – Basic Principles	5	
	3.3	FID signal-excitation & emission – Basic pulse sequences - Block diagram	5	
	Measurement of blood pressure			
4	4.1	Measurement of blood pressure – sphygmomanometer & oscillometric methods. Holter recorders.	5	4
	4.2	Photo plethysmography - for pulse rate measurement - Pulse oximeters	5	
	4.3	Cardiac stress testing – methods & protocols Patient monitoring systems-Bed side & central station	5	
5	Teacher Specific Content			

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Electronic Circuits Design			
Type of Course	SDC			
Course Code	MG2SDCIIM101			
Course Level	100			
Course Summary	This course provides a comprehensive foundation in analog electronics, focusing on the design, construction, and analysis of essential electronic circuits.. The course also emphasizes implementing signal processing applications and developing hands-on skills through various experiments involving analog circuit components and configurations.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		3	1	0
Pre-requisites, if any	NIL			

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Design and implement signal processing applications in electronic circuits	U	1,2,3,4,5,6,7,8,9,10
2	Understand the working principles of amplifiers and oscillators for electronic systems.	U	1,2,3,4,6,7,8,9,10

3	Analyze rectifiers, filters, voltage regulators, operational amplifiers, and timer circuits with their applications.	An	1,2,3,4,6,7,8,9,10
4	Able to design, construct, and analyze basic analog electronic circuits	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	1	1	1	1	2	3
CO 2	2	3	2	1	0	1	1	1	1	2
CO 3	3	3	2	1	0	1	1	1	1	2
CO 4	3	3	2	2	1	1	1	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

MGU-B.VOC. (HONOURS)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Wave shaping and multivibrator circuits			
	1.1	RC & RL Integrator and Differentiator circuits.	2	1
	1.2	Diode clippers, Diode Clampers.	4	
	1.3	Astable multivibrator, Monostable multivibrator.	4	1
1	1.1	RC & RL Integrator and Differentiator circuits.	2	1
	1.2	Diode clippers, Diode Clampers.	4	1
	1.3	Astable multivibrator, Monostable multivibrator.	4	1

	1.4	Bistable multivibrators- Schmitt trigger circuit.	5	1
2	Amplifiers and Oscillators			
	2.1	Amplifier- definition, faithful amplification, classification based on configuration, power, and frequency	5	2
	2.2	Feedback- definition, types, advantages and disadvantages, applications..	3	2
	2.3	Oscillators- definition, working, applications.	3	2
	2.4	Types of oscillators, RC phase shift and crystal oscillator- working, applications.	4	2
	Rectifiers, filters,regulator,OP AMP &555 TIMER			
3	3.1	Rectifiers- definition, half wave, centre tapped and bridge full wave rectifier-efficiency, ripple factor, PIV, ratings.	2	3
	3.2	Filters- definition Working- low pass, high pass passive and active filters, applications Regulator-definition, working	5	3
	3.3	OPAMP– definition, block diagram, operation, characteristics, applications inverting, integrator, differentiator.	4	3
	3.4	Timers– block diagram, pin diagram of 555. Multi-vibrators- Astable and monostable using 555	4	3
4	(Practical Session)			
	Lab Experiments			

4	4.1	1. To study and analyze clipping circuits and observe the waveform for series and parallel clipper using diodes. 2. To design and analyze clamping circuits and observe how the DC level of a waveform is shifted. 3. To design an op-amp integrator and observe the output waveform for various input signals. 4. To design an op-amp differentiator and study its response to square and triangular waves. 5. To construct an astable multivibrator and observe the square wave generation without any external triggering.	10	4
	4.2	6. To construct an astable multivibrator and observe the square wave generation without any external triggering. 7. To construct and study the working of an RC phase shift oscillator. 8. To construct and compare half-wave, center-tapped, and bridge rectifier circuits. 9. To study the voltage regulation property of a Zener diode in reverse bias. 10. To construct and test an inverting amplifier using 741 OPAMP. 11. To design an astable multivibrator using 555 timer and measure output frequency. 12. To design an monostable multivibrator using 555 timer and measure output frequency	20	4
5	Teacher Specific Content			



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Smart Sensors			
Type of Course	SDC			
Course Code	MG2SDCIIM102			
Course Level	100			
Course Summary	Understand the transducers, sensors and their different classifications different types of transducers with respect to the process variables, introduction to commonly used instruments for each process variables.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		3	1	
			OJT	
			0	75
Pre-requisites, if any	NIL			

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the classification, characteristics, and selection of transducers and sensors used in measurement systems.	U	1,2,3,4,6, 7,8,9,10

2	Gain knowledge of various temperature sensing elements, their principles, measuring circuits, and applications.	U	1,2,3,4,6,7,8,9,10
3	Understand different methods and devices for measuring pressure, including low-pressure techniques and solid-state sensors.	An	1,2,3,4,6,7,8,9,10
4	Familiarize and apply the smart sensors and analyze the results	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	2	3	2	1	0	1	1	1	1	2
CO 3	2	3	2	1	0	1	1	1	1	2
CO 4	3	3	2	2	1	1	1	1	2	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Transducers & sensors			
	1.1	classification of transducers primary and secondary transducers	2	1
	1.2	passive and active transducers,	4	1
	1.3	analog and digital transducers characteristics	4	1
	1.4	transfer characteristics, transfer function, error, transducer response output characteristics	5	1

2	Temperature Measurement			
	2.1	Thermocouples and its applications, RTD, thermistor, semiconductor sensors, Pyrometry, Principles, measuring circuits, characteristics, applications. based on configuration, power, and frequency	5	2
	2.2	RTD and its applications	3	2
	2.3	Thermistor-Characteristics and its applications.	3	2
	2.4	semiconductor sensors- working, applications.	4	2
	Pressure measurements and Flow measurement			
3	3.1	Elastic type, strain gauge, capacitive, inductive, , Solid state pressure gauges., Electromagnetic	2	3
	3.2	Measurement of low pressure, McLeod gauge, Thermal conductivity gauges, Ionization gauge,	5	3
	3.3	pH meters, types of electrodes	2	3
	3.4	Differential pressure type, variable area type, Rotameters	4	3
	3.5	Ultrasonic type, Measurement of humidity and moisture content	2	3
	(Practical Session)			
	Lab Experiments			

4	4.1	1. How can you distinguish between a primary and secondary transducer using a simple experimental setup? Example: Use a Bourdon tube connected to an LVDT sensor to demonstrate both types.	10	4
	4.2	2. Design a simple circuit to observe the response of a thermistor and explain how its resistance changes with temperature. 3. Using an RTD (Resistance Temperature Detector), how can you measure the temperature of water and calibrate the reading with a standard thermometer?	10	4
	4.3	4. Demonstrate the difference between active and passive transducers using a thermocouple (active) and an RTD (passive). 5. Using a digital multimeter and analog pointer meter, compare readings from a thermocouple and classify them as analog or digital transducers. 6. Plot the transfer characteristics of a thermistor sensor by measuring voltage across it at different temperatures.	10	4

		7. Measure and compare atmospheric pressure using a capacitive pressure transducer and a mechanical (elastic type) gauge. 8. Construct a basic circuit using a semiconductor temperature sensor (e.g., LM35) and measure human body temperature. 9. Using a pH meter, measure the pH of various household liquids (e.g., lemon juice, soap water) and compare readings from different electrode types.		
5	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1.Lab involvement 2.Record 3.Viva
Assessment Types	

	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks
--	---

REFERENCES :

1. “Robert L Boylestead and Louis Nashelsky”, “*Electronic Devices and circuit theory*”, Pearson, Tenth edition 2009
2. “S. Salivahanan, N. Suresh Kumar and A. Vallava Raj”, “*Electronic Devices and circuits*”, TMH, 2nd Edition 2008.
3. David A. Bell, “*Electronic Devices and Circuits*”, Fifth Edition, Oxford University Press, 2008.

SUGGESTED READINGS:

1. Millman J., Halkias C.C., Jit S., “*Electronic Devices and Circuits*”, Tata McGraw-Hill, 2nd Edition.
2. John G. Proakis & Masoud Salehi, “*Communication System Engineering*”, 2nd Edition, 2002.
3. Kennedy G., Davis B., “*Electronic Communication Systems*”, Tata McGraw-Hill, 4th Edition.
4. Sanjay Sharma, “*Communication Systems, Analog & Digital*”, S.K. Kataria & Sons, 5th Edition, 2009.

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	3-Dimensional Printing Technology			
Type of Course	MDC			
Course Code	MG2MDCIIM100			
Course Level	100			
Course Summary	To equip learners with foundational knowledge and practical skills in 3D printing technologies and Arduino-based robotics, enabling them to design, prototype, and implement innovative solutions, and to effectively guide others through hands-on instruction and application.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		2	1	0
Pre-requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the fundamentals of 3D printing technologies, materials, and processes, and apply this knowledge for design, prototyping, and innovative applications.	U	1,2,3, 4,5,6, 7,8,9, 10

2	Understand the basic principles, types, and working mechanisms of various 3D printing technologies such as FDM, SLA, and SLS..	A	1,2,3, 4,5,6, 7,8,9, 10
3	Identify and select appropriate 3D printing materials based on properties like strength, flexibility, and application requirements.	U	1,2,3, 4,5,6, 7,8,9, 10
4	Use basic CAD software to design 3D models and prepare them for printing through slicing tools and printer setup for simple prototypes.	U	1,2,3, 4,5,6, 7,8,9, 10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	3	1	1	1	2	1	2	3
CO 2	3	3	3	1	1	1	1	1	2	3
CO 3	3	3	3	2	2	1	1	1	3	3
CO 4	3	2	2	3	3	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	3D Printing			
	1.1	Introduction to 3D Printing: Definition and evolution of 3D Printing/Additive Manufacturing (AM)	3	1
	1.2	Materials for 3D Printing: Thermoplastics (PLA, ABS, PETG), Photopolymers (resins for SLA/DLP)	4	1
	1.3	Future Trends and Challenges: Sustainability and recyclability in 3D printing 4D Printing and smart materials.	3	1

2	Introduction to 3D Printing			
	2.1	Applications across industries (aerospace, medical, consumer goods) Advantages and limitations of 3D printing Types of 3D Printing Technologies Working principles, process flow, and components of each type Comparison of printing techniques: speed, resolution, cost, and usage Practical Prototyping and Post-Processing Printer setup and calibration Printing a basic part (hands-on activity) Troubleshooting common issues (warping, stringing, under-extrusion) Post-processing: support removal, sanding, painting.	7	2
	2.2	Materials for 3D Printing Resins, powders, metal filaments Material properties and selection criteria for various applications Environmental and safety considerations	5	2
	2.3	CAD Design and Slicing for 3D Printing	5	2

		Basics of 3D design using CAD software (TinkerCAD/Fusion 360)		
	2.4	Model optimization for 3D printing	3	2
	Practical Session			
	Lab experiments			
3	3.1	1. Introduction to 3D Printer and Slicing Software. 2. Design and Print a Basic 3D Object. 3. Testing Material Properties.	15	3
	3.2	4. To model and print a prototype (e.g., phone stand, bottle holder). 5. Error Analysis and Troubleshooting.	15	4
4	Teacher Specific Content			

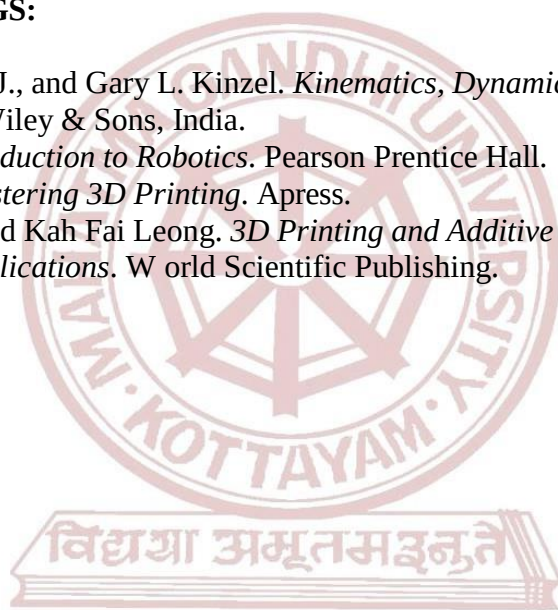
Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical(25 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination (50 Marks, 1.5 hrs) Part A : Short Answer Questions(5 out of 7 Questions) - (5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20Marks) Record : 10 Marks

REFERENCES:

1. Blum, Jeremy. *Exploring Arduino: Tools and Techniques for Engineering Wizardry*. Paperback ed., 9 Aug. 2013.
2. Gibson, Ian, David Rosen, and Brent Stucker. *Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing*. 2023.

SUGGESTED READINGS:

1. Waldron, Kenneth J., and Gary L. Kinzel. *Kinematics, Dynamics & Design of Machinery*. John Wiley & Sons, India.
2. Craig, John J. *Introduction to Robotics*. Pearson Prentice Hall.
3. Horvath, Joan. *Mastering 3D Printing*. Apress.
4. Chua, Chee Kai, and Kah Fai Leong. *3D Printing and Additive Manufacturing: Principles and Applications*. World Scientific Publishing.



MGU-B.VOC. (HONOURS)

Syllabus



SEMESTER III

MGU-B.VOC. (HONOURS)

Syllabus

Semester: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG3SDCIIM200	Industrial Instruments	SDC	4	4	4	0	0
MG3SDCIIM201	Printed Circuit Board Designing and Simulation	SDC	4	5	3	2	0
MG3SDCIIM202	Programmable Logic Devices	SDC	4	5	3	2	0
MG3MPCIIM200	Measurement Technology	MPC	4	5	3	2	0
MG3MDCIIM200	E-Waste Management systems in Kerala	MDC	3	3	3	0	0
MG3SDCIIM203	On-the-Job Training	SDC	2	5	0	0	5

L — Lecture, P — Practical/Practicum , O — On the Job Training



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Industrial Instruments				
Type of Course	SDC				
Course Code	MG3SDCIIM200				
Course Level	200				
Course Summary	To understand the instruments and their different classifications with respect to the process variables, introduction to commonly used instruments for each process variable.				
Semester	3	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	
Pre- requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand and explain the working of various level measurement systems for different applications.	U	1,2,3, 4,6,7, 8,9,10

2	Explain the principles, construction, and applications of vacuum measuring instruments and pumps, and apply knowledge of gauges and vacuum systems in industrial settings.	A	1,2,3, 4,5,6, 7,8,9, 10
3	Describe the working of pH measurement systems using different electrodes, especially the glass electrode, and understand its role in chemical industries.	U	1,2,3, 4,6,7, 8,9,1 0
4	Understand ,identify and explain the structure and function of smart transmitters, modern sensor technologies safety-related detectors used in industrial environments.	U	1,2,3, 4,5,6, 7,8,9, 10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	3	3	2	1	1	2	1	1	2	3
CO 3	2	3	2	1	0	2	2	1	1	2
CO 4	3	3	3	2	1	2	2	2	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Level Measurement			
	1.1	Capacitance probe; conductivity probes	5	1
	1.2	Diaphragm level detector, differential pressure level detector	6	1
	1.3	Radiation level sensors, RADAR level gauges, level transmitter, ultrasonic level detector	4	1

2	Vacuum measurement			
	2.1	Vacuum measurement and applications	4	2
	2.2	Vacuum measuring instruments- gauges, pirani gauge (principle, construction and working, advantages and disadvantages)	3	2
	2.3	Pumps- rotary pumps, root blowers (principle, construction and working, advantages and disadvantages)	8	2
3	3.1	pH - Definition, types of electrodes	4	3
	3.2	glass electrode pH measurement,	6	
	3.3	Application in Chemical industries (principle, construction and working, advantages, disadvantages)	5	
4	4.1	Smart transmitter - Block diagram-, Recent trends in sensor technology.	5	4
	4.2	Recent trends in sensor technology, Semiconductor sensors, Film sensors, MEMS	5	
	4.3	Nanosensors (principle, construction and working, advantages, disadvantages). Detectors - Smoke detectors, LPG detectors	5	
5	Teacher specific content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Theory(30 Marks) 1. Written Test 2. Problem Based Assignment 3. Seminars
	B. End Semester Evaluation (ESE) ESE for Theory : Written Test(70 Marks , 2 hrs) Part A: Very Short Answer Questions (Answer all)- (10*2=20 Marks) Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C:Essay Questions(2 out of 3 Questions) - (2*15=30 Marks)

REFERENCES:

1. Sawhney, A. K., and Puneet Sawhney. *A Course in Electrical and Electronics Measurements and Instrumentation*. Dhanpat Rai & Co., New Delhi, India, latest ed. Print.
2. Sawhney, A. K., and Puneet Sawhney. *A Course in Mechanical Measurements and Instrumentation & Control*. Dhanpat Rai & Co., New Delhi, India, latest ed. Print.

SUGGESTED READINGS:

1. McMillan, Gregory K., and Douglas M. Considine, editors. *Process/Industrial Instruments and Controls Handbook*. 6th ed., McGraw-Hill Education, New York, NY, 2014. Print.
2. Singh, S. K. *Industrial Instrumentation and Control*. McGraw-Hill Education, New Delhi, India, 2012. Print.
3. Doebelin, Ernest O. *Measurement Systems: Application and Design*. 6th ed., McGraw-Hill Education, New York, NY, 2017. Print.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Printed Circuit Board Designing and Simulation			
Type of Course	SDC			
Course Code	MG3SDCIIM201			
Course Level	200			
Course Summary	This course is to enable students for designing PCB using software. PCB (Printed Circuit Board) designing is an integral part of each electronics products and this program is designed to make students capable to design their own projects PCB up to industrial grade.			
Semester	3	Credits		4
Course Details	Learning Approach	Lecture	Practical	OJT
		3	1	0
Pre- requisites, if any	NIL			
				Total Hours
				75

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Students will learn the basics of PCB design and simulation using software tools.	U	1,2,3,4,6,7,8,9,10
2	Students will learn about different types of electronic component packages used in PCB design.	U	1,2,3,4,6,7,8,9,10
3	Students will understand various PCB layers, board design elements, and commonly used PCB materials..	An	1,2,3,4,6,7,8,9,10
4	Gain hands-on experience in circuit simulation, schematic design, and PCB layout using tools like Proteus and understand the PCB manufacturing process.	A	1,2,3,4,5,6,7,8,9,10

***Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)**

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	3
CO 2	2	2	2	1	0	1	1	1	1	2
CO 3	2	2	2	1	0	1	2	1	1	2
CO 4	3	3	3	2	1	2	2	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Basics of PCB and Circuit Simulation			
	1.1	Introduction & Brief History What is PCB, Difference between PWB and PCB.	2	1
	1.2	Types of PCBs – Single Sided (Single Layer) Multi-Layer (Double Layer), PCB Materials	4	1
	1.3	Circuit Simulation using P-Spice Introduction, Description of P-Spice, Types of analysis	4	1
	1.4	Description of simulation software tools (like OrCAD / PROTEL / Proteus / Microcap / Eagle)	5	1
	Component Package types			
	2.1	Component Package Types – Through Hole Packages – Axial lead , Radial Lead	5	2
	2.2	Single Inline Package (SIP), Dual Inline Package (DIP), Transistor Outline (TO)	3	2

2	2.3	Through Hole Packages – Metal Electrode Face (MELF), Leadless Chip Carrier (LCC), Small Outline Integrated Circuit (SOIC).	3	2
	2.4	Quad Flat Pack (QFP) , Thin QFP (TQFP), Ball Grid Array (BGA), Plastic Leaded Chip Carrier (PLCC)	4	
PCB Layers				
3	3.1	PCB Layers – Electrical Layers, Top Layer, Mid Layer, Bottom Layer Mechanical Layers.	2	3
	3.2	Board Outlines , Cut-outs, Drill Details, Documentation Layers	5	3
	3.3	Components Outlines, Reference Designation, Text.	4	3
	3.4	PCB Materials – Standard FR-4 Epoxy Glass, Multifunctional FR-4, Tetra Functional FR-4, Polyimide Glass, Teflon.	4	3
	(Practical Session)			
	Lab Experiments using proteus software			
	4.1	1. Simulate Ohm's Law: Design a basic resistor circuit and verify voltage/current relationship. 2. Verify KVL and KCL using proteus simulation software. 3. Simulate a simple diode rectifier circuit (half-wave) using proteus. 4. Simulate a simple diode rectifier circuit (full wave) using proteus.	10	4

4	4.2	5. Design RC differentiator circuit using proteus simulation. 6. Design RC integrator circuit using proteus simulation. 7. Simulate astable multivibrator using proteus simulation. 8. Draw and simulate RC coupled Amplifier.	20	4
		9. Draw and simulate RC phase shift oscillator. 10. Design an inverting op-amp circuit and observe output waveform. 11. Create and simulate a logic gate circuit (AND, OR, NOT). 12. Simulate a 555 Timer in Astable mode (LED blinking or pulse generation).		
5	 Teacher Specific Content			

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc (Honours) Industrial Instrumentation and Automation			
Course Name	Programmable Logic Devices			
Type of Course	SDC			
Course Code	MG3SDCIIM202			
Course Level	200			
Course Summary	This course provides a comprehensive introduction to microprocessors and microcontrollers, their architecture, programming, and applications in embedded systems. It focuses on understanding the internal working of microprocessors (like Intel 8085/8086) and microcontrollers (like 8051), including instruction sets, interfacing techniques, and control applications.			
Semester	3	Credits		4
Course Details	Learning Approach	Lecture	Practical	OJT
		3	1	0
Pre- requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the architecture and functional blocks of the 8085 microprocessor, including the roles of its registers, ALU, and control unit.	U	1,2,3,10
2	Differentiate between microprocessors, microcontrollers, and embedded processors	An	1,2,3,4,10
3	Understand and explain the structure of a C program, use of variables, identifiers, constants, literals, and built-in data types.	U	1,2,3,4,10
4	Develop and simulate programs in KEIL to perform programming	A	1,2,3,4,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	1	0	0	0	0	0	0	2
CO 2	3	3	2	1	0	0	0	0	0	2
CO 3	3	3	1	2	0	0	0	0	0	2
CO 4	3	3	1	2	0	0	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Microprocessor			
	1.1	8085 Microprocessor Architecture, Address, Data and Control Buses, 8085 Pin Functions, De multiplexing of Buses,	5	1
	1.2	Generation Of Control Signals, Instruction Cycle, Machine Cycles, T-States, Memory Interfacing, Classification of Instructions	5	1
	1.3	Addressing Modes, 8085 Instruction Set, simple programs ang simulations, Peripheral IC- 8255, 8279, 8253, 8257	5	1
2	Introduction to Microcontrollers and Embedded Processors			
	2.1	8051 Microcontrollers Hardware - Microcontroller Architecture	5	2
	2.2	IO Port structure, memory organization, Special Function Registers (SFRs).	5	2
	2.3	- Instruction Set, addressing modes, Instruction Types, Serial Communication	5	2
3	Introduction to C.			
	3.1	Variables and identifiers, built-in data types. Variable definition, Arithmetic operators and expressions, constants and literals	5	3
	3.2	Variables and identifiers, built-in data types. Variable definition, Arithmetic operators and expressions, constants and literals	5	3

	3.3	Statement, basic input/output statement, simple 'c' programs, conditions, relational operators, logical connectives, if statement, if-else statement, loops	5	3
4	Introduction to KEIL			
	4.1	1. Develop a program to Add two numbers 2. Develop a program to subtract two numbers	10	4
	4.2	3. To generate Fibonacci series 4. Numbers arrangement in ascending order	10	4
	4.3	5. Numbers arrangements in descending order 6. seven segment keyboard interfacing	10	4
5	Teacher Specific Content			



MGU-B.VOC. (HONOURS)

Syllabus

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT Mode of Assessment
Assessment Types	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES:

1. Mazidi, Muhammad Ali, Rolin D. McKinlay, and Janice Gillispie Mazidi. *The 8051 Microcontroller and Embedded Systems: Using Assembly and C*. 2nd ed., Pearson Education, Upper Saddle River, NJ, 2006. Print.
2. Ayala, Kenneth J. *The 8051 Microcontroller: Architecture, Programming, and Applications*. 3rd ed., Cengage Learning, Boston, MA, 2004. Print.

SUGGESTED READINGS:

1. Gaonkar, Ramesh S. *Microprocessor Architecture, Programming, and Applications with the 8085*. 6th ed., Penram International Publishing, Mumbai, 2011. Print.
2. Ram, B. *Fundamentals of Microprocessors and Microcomputers*. Dhanpat Rai Publications, New Delhi, latest ed. Print.



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Subject	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Measurement Technology			
Type of Course	MPC			
Course Code	MG3MPCIIM200			
Course Level	200			
Course Summary	Provides foundational knowledge and practical skills in the measurement and control of physical variables in industrial and laboratory settings. This course introduces the principles, devices, and techniques used to measure parameters such as temperature, pressure, flow, level, and electrical quantities.			
Semester	3	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		3	1	
			OJT	
			0	75
Pre- requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify different types of proximity sensors	U	1,2,3,4,5,9,10
2	Understand characteristics of Vacuum System	U	1,2,3,4,8,9,10
3	Understand the working of pH detectors	U	1,2,3,4,8,9,10
4	.Familiarize with handheld tools	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/P O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	2	1	1	0	0	0	2	1
CO 2	3	3	2	1	0	0	0	1	1	2
CO 3	3	3	2	1	0	0	0	1	2	2
CO 4	2	2	3	2	1	1	1	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Module	Units	Course description	Hrs	CO No.
1	Proximity Sensors			
	1.1	Capacitance probe; conductivity probes	5	1
	1.2	Diaphragm level detector, differential pressure level detector	5	1
	1.3	Radiation level sensors, RADAR level gauges, level transmitter, ultrasonic level detector	5	1
	Vacuum measurement			
2	2.1	Vacuum measurement and applications	5	2
	2.2	Vacuum measuring instruments- gauges, pirani gauge (principle, construction and working, advantages and disadvantages)	5	2
	2.3	Pumps- rotary pumps, root blowers (principle, construction and working, advantages and disadvantages)	5	2
	pH measurement			
3	3.1	pH - Definition, types of electrodes	5	3
	3.2	glass electrode pH measurement,	5	3
	3.3	Application in Chemical industries (principle, construction and working, advantages, disadvantages)	5	3
Practical Session				

4	4.1	1.Familiarize handheld tools with neat figures 2.Measure diameter Meter of given work piece using Vernier 3. Measure diameter Meter of given work piece using screw gauge	10	4
	4.2	4.Drill a hole on the given wooden piece and measure depth & diameter Meter 5.Use hand drill to make a groove and measure depth 6. Speed measurement by using Tachometer	10	4
	4.3	7.Distance measurement using ultrasonic meter 8.Angle measurement using clinometers 9.Familiarize CRO	10	4
5	Teacher Specific Content			



MGU-B.VOC. (HONOURS)

Syllabus

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT Mode of Assessment
Assessment Types	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES: MGU-B.VOC. (HONOURS)

1. Albert D. Helfrick *Modern electronics Instrumentation and measurement techniques*
2. Umesh Sinha, Satyaprakashan, *Electrical and Electronics measurements and instrumentation*, Tech India publication 1992

SUGGESTED READINGS :

1. R.K. Rajput, *Basic mechanical engineering*, Laxmi Publications
2. R.S.Khurmi,J.K.Gupta, *Work shop technology*, S.Chand publishers



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	E-Waste Management systems in Kerala				
Type of Course	MDC				
Course Code	MG3MDCIIM200				
Course Level	200				
Course Summary	This course addresses the imperative for sustainable practices in Electronics. By instilling an understanding of eco- friendly principles, providing hands on experience in E-waste management fostering critical thinking and sustainability consciousness..				
Semester	3	Credits			Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	0	0	
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

Syllabus

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Explain The threat of E-waste on human health and the environment.	U	1,2,3,4,5,6,7, 8,9,10
2	Construct knowledge sustainable materials for electronic devices	A	1,2,3,4,5,6,7, 8,9,10
3	Develop E-waste management practices and strategies for recycling electronic products	U	1,2,3,4,5,6,7, 8,9,10

4	Apply green electronics principles to real world scenarios and obtain a fundamental understanding of future trends of green Electronics	U	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	2	1	1	1	2	2	3
CO 2	2	3	1	1	1	1	1	2	1	2
CO 3	3	3	2	2	2	1	1	2	3	2
CO 4	3	3	2	2	2	1	1	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Sources of E-Waste			
	1.1	Electronic waste (E-waste), sources of E-Waste, categories, Effect of E-waste on ecosystems.	5	1
	1.2	Health hazards of E-waste. Challenges associated with the disposal of E-waste. Benefits of E-waste recycling	6	1
	1.3	E-waste compounds and its toxicity chart	4	1
2	Sustainable materials			
	2.1	Definition and significance of sustainable materials in the context of electronic device	2	2
		Different categories of		

	2.2	sustainable materials:(Recycling of copper, aluminum, gold from PCB	5	2
	2.3	Eco friendly dielectric layers - Paper, Silk, cellulose and cellulose derivatives, Resin, Gelatin, Shellac, Organic semiconductor materials	5	2
	2.4	Performance and durability of sustainable materials compared to traditional ones.	3	2
	Recycling Methods			
3	3.1	Mechanical Recycling Methods Introduction to mechanical recycling Shredding, Magnetic Separation, Air Classification, Gravity Separation	5	3,4
	3.2	Chemical Processes for Material Recovery- Leaching, Solvent, Extraction, Pyrolysis, Electrochemical Processes	5	
	3.3	Case study - Identification and separation of reusable components inside a PC	5	
4	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) ● Use of ICT tools in conjunction with traditional classroom teaching methods ● Interactive sessions ● Class discussions
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical(25 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination (50 Marks, 1.5 hrs) Part A : Short Answer Questions(5 out of 7 Questions) - (5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20Marks) Record : 10 Marks

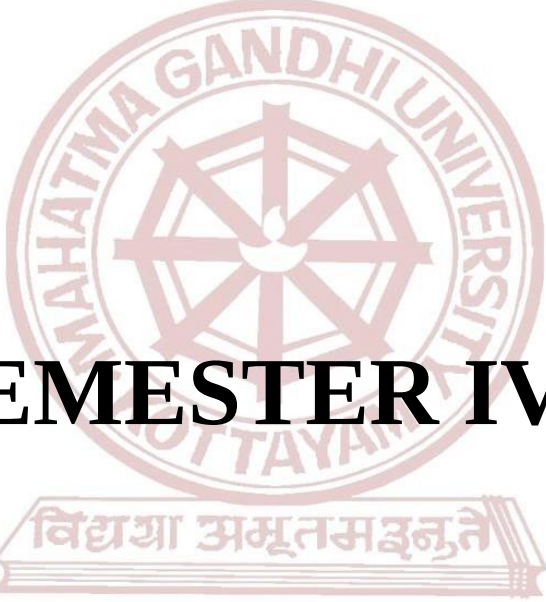
MGU-B.VOC. (HONOURS)

REFERENCES:

1. Erach Bharucha – *Textbook of Environmental Studies for Undergraduate Courses*
2. R. Rajagopalan – *Environmental Studies: From Crisis to Cure*
3. Kaushik, C.P. & Kaushik, Anubha – *Perspectives in Environmental Studies*

SUGGESTED READINGS:

1. Sharma, P. D. – *Ecology and Environment*
2. <https://moef.gov.in/>
3. <https://cpcb.nic.in/>

The logo of Mahatma Gandhi University is a circular emblem. It features a central wheel with eight spokes, resembling a Dharma Chakra, with a small flame at its center. The words "MAHATMA GANDHI UNIVERSITY" are inscribed in a circular border around the wheel. Below the wheel, there is a banner with the Sanskrit motto "विद्यया अमृतमश्नुते" (Vidyayā amṛtam aśnute).

SEMESTER IV

MGU-B.VOC. (HONOURS)

Syllabus

Semester: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours/ Week	Hour Distribution /week		
					L	P	O
MG4SDCIIM200	Fibre Optic Technology	SDC	4	4	4	0	0
MG4SDCIIM201	Final Control Elements	SDC	4	5	3	2	0
MG4SDCIIM202	Embedded Automation	SDC	4	5	3	2	0
MG4MPCIIM200	Industrial Robotics	MPC	4	5	3	2	0
MG4SECIIM200	Information Technology (IT) for entrepreneurs	SEC	3	3	3	0	0
MG4VACIIM200	Environmental studies for Instrumentation	VAC	3	3	3	0	0
MG4INTIIM200	Internship	INT	2	60 hours			

L — Lecture, P — Practical/Practicum , O — On the Job Training



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Fibre Optic Technology				
Type of Course	SDC				
Course Code	MG4SDCIIM200				
Course Level	200				
Course Summary	To understand the instruments and their different classifications with respect to the process variables, introduction to commonly used instruments for each process variable.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	
Pre- requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Students will gain a comprehensive understanding of the nature and behavior of light and electromagnetic waves. They will explore the principles of diffraction and interference, and understand how electromagnetic waves.	U	1,2,3,4,6,7,8,9, 10

2	Upon completion of this course, students will be able to understand the characteristics and working principles of various light sources including natural, incandescent, gas discharge lamps, LEDs, infrared sources, and semiconductor lasers.	A	1,2,3, 4,6,7, 8,9,10
3	The student will gain foundational knowledge of optical fibers, including their types, classifications, and applications for fiber characterization and fault detection in optical communication systems.	U	1,2,3, 4,5,6, 7,8,9, 10
4	Students will be able to explain the working principles and characteristics of different light sources such as natural light, incandescent lamps, gas discharge lamps, LEDs, infrared sources, and semiconductor lasers.	An	1,2,3, 4,6,7, 8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	3
CO 2	2	3	2	1	0	1	1	1	1	2
CO 3	3	3	2	1	1	1	2	1	2	3
CO 4	2	3	2	1	0	1	1	1	1	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT MGU-B.VOC. (HONOURS)

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Nature of Light			
	1.1	electromagnetic optics spectrum, propagation of light, ,	5	1
	1.2	electromagnetic waves in dielectric media,	6	1
	1.3	absorption, scattering, dispersion, diffraction and interference.	4	1
	Sources of Light			

2	2.1	Light sources, natural sources,	4	2
	2.2	incandescent lamp, gas discharge lamp.	3	2
	2.3	Light-emitting diodes- LED structures, infrared sources, semiconductor laser, absorption, spontaneous emission and stimulated emission	8	2
	Fundamental of Fibers			
3	3.1	Fundamental of Fibers, single mode fibers, types and classification of fibers, Attenuation,	4	3
	3.2	Optical Time Domain Reflectometer(OTDR) Introduction to OTDR, Working Principle of OTDR, Applications of OTDR	6	
	3.3	Block Diagram of OTDR, Pulse Reflection and Backscatter Measurement, Types of OTDR (Conventional, High-Resolution, etc.), Applications of OTDR Types of OTDR (Conventional, High-Resolution, etc)	7	
4	4.1	Light dependent resistors (LDR), - principle, working ,characteristics, general applications	8	4
	4.2	Application of LDR in advanced technology	5	
5	Teacher specific content			

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Final Control Elements			
Type of Course	SDC			
Course Code	MG4SDCIIM201			
Course Level	200			
Course Summary	This course provides comprehensive knowledge on the design, operation, and selection of valves and actuators, essential components for controlling the flow of fluids, gases, or slurries in various industrial systems. It covers their types, functions, applications, and troubleshooting methods.			
Semester	4	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		3	1	0
Pre- requisites, if any	NIL			

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify and describe the basic parts of a valve and their functions.	U	1,2,3,10
2	Interpret technical specifications, symbols, and standards related to valves	An	1,2,3,4,8,10
3	Apply standard practices for safe maintenance and troubleshooting of actuators.	A	1,2,3,4,5,6,7,8,9,10
4	Demonstrate safe and correct procedures for disassembling valves.	A	1,2,3,4,5,6,7,8,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	1	1	0	0	0	0	0	0	2
CO 2	3	3	2	1	0	0	0	1	0	2
CO 3	3	3	1	1	1	1	1	1	1	2
CO 4	2	2	1	1	1	1	1	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Valve-Parts			
	1.1	Principle of operation, Characteristics of valve	5	1
	1.2	Drawings	5	1
	1.3	Body, Bonnet, ports, handle, actuator, disc stem, gasket, spring, Trim, spring	5	1
2	Various types of valves			
	2.1	Gate, expanding gate, plug, ball, check, bleed	5	2
	2.2	Drawings	5	2
	2.3	Butterfly valve, Globe valve, Disk check valve	5	2
3	Introduction to actuators			
	3.1	Basic actuators and actuator operation, classification reverse & direct actuators	5	3
	3.2	Pneumatic, hydraulic & Electric actuators.	5	3
	3.3	Maintaining actuators safely, Configuration of control elements	5	3
4	Valve assembly			
	4.1	1.To identify & draw Globe valve 2. To identify & draw Ball valve To identify & draw Gate valve	10	4

	4.2	4. To identify & draw Globe valves. 5. To identify need & draw valves To identify & draw check valves.	10	4
	4.3	7. Dismantle the given control valve Reassemble the given control valve	10	4
5	Teacher Specific Content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction)
	• Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT
	Mode of Assessment
Assessment Types	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

REFERENCES:

1. Lipták, Béla G., editor. *Flow Measurement*. CRC Press, Boca Raton, FL, 2016. Print.
2. Lipták, Béla G., editor. *Instrument Engineers' Handbook: Process Measurement and Analysis*. 4th ed., CRC Press, Boca Raton, FL, 2003. Print.

SUGGESTED READINGS:

1. Skousen, Philip L. *Valve Handbook*. 3rd ed., McGraw-Hill Education, New York, NY, 2011. Print.
2. Johnson, Curtis D. *Process Control Instrumentation Technology*. 8th ed., Pearson Education, Upper Saddle River, NJ, 2013. Print.
3. Perry, Robert H., and Don W. Green, editors. *Perry's Chemical Engineers' Handbook*. 9th ed., McGraw-Hill Education, New York, NY, 2019. Print.
4. Considine, Douglas M., editor. *Process Instruments and Controls Handbook*. 5th ed., McGraw-Hill Education, New York, NY, 1999. Print.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Embedded Automation				
Type of Course	SDC				
Course Code	MG4SDCIIM202				
Course Level	200				
Course Summary	The objective of this course is to teach students design and interfacing of microcontroller-based embedded systems. High-level languages are used to interface the microcontrollers to various applications. There are extensive hands-on labs/projects. Embedded system for sensor applications will be introduced.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	
Pre-requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basics of PIC microcontrollers, their advantages, types, products, and internal architecture.	U	1,2,3,4,6,7,8,9,10

2	Learn the basics of AVR architecture, pipelining features and register structure.	U	1,2,3,4,6,7,8,9,10
3	Learn to interface and program display units, handle interrupts, implement communication protocols with real-time applications.	An	1,2,3,4,5,6,7,8,9,10
4	Learn to write and execute basic ARM assembly language programs for arithmetic operations and logical problem solving.	A	1,2,3,4,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	2	3	2	1	0	1	1	1	1	2
CO 3	3	3	3	2	1	2	1	1	2	3
CO 4	3	3	2	1	0	1	1	1	1	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	PIC micro controllers.			
	1.1	Introduction to PIC micro controllers.	2	1
	1.2	Advantage of PIC micro controllers	4	1
	1.3	Types and products of PIC micro controllers	4	1
	1.4	Architecture of PIC micro controllers	5	1
	AVR microcontrollers			
	2.1	Introduction of AVR Architecture Pipelining.	5	2

2	2.2	Digital Basics, Pin configuration of ATMEGA16	3	2
	2.3	Register structure of ATMEGA16,32.	3	2
	2.4	AVR Compiler IO Registers in ATMEGA16, 32.	4	2
	Peripheral Devices and Communication Protocols			
3	3.1	Display Units: Types of display devices 7-segment displays: Common Anode & Common Cathode types 16x2 LCD: Pin diagram, applications	2	3
	3.2	Interrupts: Definition and types of interrupts Handling interrupts in microcontrollers	5	3
	3.3	Serial Communication: Basics and importance Hardware description: Logical level converter and MAX232	4	3
	3.4	I2C Protocol: Overview and working Advantages and disadvantages of I2C protocol	4	3
4	Practical session			
	Lab experiments using PIC and AVR microcontrollers.			

	4.1	1. Familiarisation of MPLAB software 2. Write a program in assembly language to find addition of two numbers using MPLAB software 3. Write a program in assembly language to find subtraction of two numbers using MPLAB software 4. Write a program in assembly language to find Multiplication of two numbers using MPLAB software.	10	4
	4.2	5. Led blinking pic using MPLAB software and simulate in proteus 6. Led blinking pic using switch MPLAB software and simulate in proteus 7. Square wave pic and simulate in proteus 8. Familiarisation of ATMEL studio for AVR 9. Led blinking avr and simulate in Proteus 10. square wave avr and simulate in Proteus	20	4
5	Teacher Specific Contents			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical (30 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination(70 Marks,3 hrs) Part A : Short Answer Questions (5 out of 7 Questions) – (5*4=20 Marks) Part B : Practical Questions(2 Questions) – (2*15=30 Marks) Viva : 10 Marks Record : 10 Marks

MGU-B.VOC. (HONOURS)

REFERENCES:

1. Mazidi, Muhammad Ali, Rolin D. McKinlay, and Danny Causey. *The PIC Microcontroller and Embedded Systems: Using Assembly and C for PIC18*. Prentice Hall, Upper Saddle River, NJ, 2007. Print.
2. *Design Reference Notes and Data Sheets of Microchip Devices*. Microchip Technology Inc., various years. Print.

SUGGESTED READING:

1. Peatman, John B. *Design with PIC Microcontrollers*. Pearson Education, Upper Saddle River, NJ, 2003. Print.
2. Mazidi, Muhammad Ali, and Sarmad Naimi. *AVR Microcontroller and Embedded Systems: Using Assembly and C*. Pearson Education Ltd., 2011. Print.
3. Gadre, Dhananjay V. *Programming and Customizing the AVR Microcontroller*. McGraw-Hill Education, New York, NY, 2001. Print.



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Subject	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Industrial Robotics				
Type of Course	MPC				
Course Code	MG4MPCIIM200				
Course Level	200				
Course Summary	To introduce the functional elements of Robotics. This course is designed to introduce and familiarize students with programming using the Arduino Micro-controller and controlling robots using Arduino.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basics of robotics, including types, components, and key concepts.	U	1,2,3,4,5,6,7,8,9,10
2	Understand the construction, types, and control of manipulators and grippers used in robotic systems.	U	1,2,3,4,5,6,7,8,9,10
3	Understand basic Arduino programming and sensor interfacing for building simple robotic automation systems.	An	1,2,3,4,5,6,7,8,9,10
4	Develop hands-on skills in basic Arduino programming and interfacing through simple experiments	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	2	1	1	1	2	2	3
CO 2	2	3	1	1	1	1	1	2	1	2
CO 3	3	3	2	2	2	1	1	2	3	2
CO 4	3	3	2	2	2	1	1	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Robotics- Basic concepts, power sources and sensors			
	1.1	Definition of robotics Origin of robotics	2	1
		Different types of robots		
	1.2	Degree of freedom, Asimov's laws of robotics, Dynamic stabilization of robots.	4	1
	1.3	Types of motors, Motor controllers, Control motor speed, Types of sensors.	4	1
	1.4	Nickel cadmium batteries, Path determination, Vision, Ranging, Laser	5	1
2	Manipulators, actuators and grippers			
	2.1	Construction of manipulators, Manipulator dynamics and force control.	5	2
	2.2	Electronics manipulator, pneumatic manipulator, control circuits	3	2
	2.3	Pneumatic, hydraulic, electric actuators and effectors	3	2
	2.4	various types of grippers	4	2
	Arduino Programming concept			

3	3.1	Arduino Logic Building Robotic Automation Basic programming Basic of Relay	2	3
	3.2	Arduino for Building Robotics Applications, Advantages of using Arduino	5	3
	3.3	• Concepts of Different types of sensors • Application of different sensors	4	3
	3.4	• Interfacing with single sensor • Interfacing with multiple sensors	4	3
Practical session				
Lab experiments using Arduino				
	4.1	1. Familiarization of Arduino 2. Blinking LED 3. Button Controlled LED 4. Traffic Light System	10	4

MGU-B.VOC. (HONOURS)

Syllabus

4	4.2	5. LED Fading with PWM 6. Buzzer Sound Control 7. Temperature Monitoring with LM35 Sensor 8. Light Detection using LDR 9. Ultrasonic Distance Measurement 10. Servo Motor Control 11. Potentiometer Controlled LED Brightness. 12. LCD Display with Arduino	20	4
5	Teacher Specific Contents			



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Information Technology(IT) for entrepreneurs			
Type of Course	SEC			
Course Code	MG4SECIIM200			
Course Level	200			
Course Summary	This course provides a comprehensive understanding of entrepreneurship, exploring key concepts such as the definition, traits, and role of entrepreneurs in India, including a focus on women entrepreneurs. It delves into the entrepreneurial ecosystem, highlighting institutions that support entrepreneurs.			
Semester	4	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		3	0	0
Pre-requisites, if any				

COURSE OUTCOMES (CO)

Syllabus

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Remember Venture Management , entrepreneurship, types of entrepreneurs,	R	1,2,3,6
2	Applying format, and manage documents using word processing tools	A	2,4,10
3	Applying and present data and information effectively using spreadsheet	A	2,3,4,8,10

4	Applying the features of Ms-PowerPoint	A	3,4,5,6,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	0	0	2	0	0	0	0
CO 2	0	3	0	3	0	0	0	0	0	2
CO 3	0	3	2	3	0	0	0	2	0	2
CO4	0	0	3	3	2	3	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Entrepreneurship			
	1.1	Entrepreneur and Entrepreneurship: Definition and Traits	4	1
	1.2	Classification of Entrepreneurs	3	1
	1.3	Women Entrepreneurs	2	1
	1.4	Role of Entrepreneurs in India	2	1
	1.5	Entrepreneurial Development Programme (EDP): Concept and Need for Training, Phases of EDP, Curriculum & Contents of Training Programme	4	1
	Word Processing Package:			

2	2.1	Introduction - Creating new Documents; Basic Editing, Saving a Document;	3	2
	2.2	Printing a Document; Print Preview, Page Orientation - Viewing Documents;	4	2
	2.3	Setting tabs - Page Margins; Indents; Ruler, Formatting Techniques; Font Formatting, Paragraph Formatting; Page Setup;	3	3
	2.4	Headers & Footers; Bullets and Numbered List; Borders and Shading; Find and Replace; Page Break & Page ;	2	3
	2.5	Numbers; Mail Merging-Spelling and Grammar Checking; Tables; Formatting Tables	3	3
3	Spreadsheet Package and Presentation Package:			
	3.1	Introduction, Excel User Interface, Working with cell and cell addresses, Selecting a Range,	2	4
	3.2	Moving, Cutting, Copying with Paste, Inserting and Deleting cells, Freezing cells, Adding,	3	4
	3.3	Deleting and Copying Worksheet within a workbook, Renaming a Worksheet. Cell Formatting Options,	3	4
	3.4	Formatting fonts, Aligning, Wrapping and Rotating text, Using Borders, Boxes and Colors, Centering a heading Insert Comments, Clear contents in a cell. Using print Preview, Margin and Centering a Worksheet, Using header and footer	3	4
	3.5	Ms-PowerPoint: Advantages of Presentation Screen layout creating presentation inserting slides adding sounds & videos-formatting slides	4	4
5	Teacher Specific Contents			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical(25 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination (50 Marks, 1.5 hrs) Part A : Short Answer Questions(5 out of 7 Questions) - (5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20Marks) Record : 10 Marks

REFERENCES

1. Drucker, Peter. *Innovation and Entrepreneurship*. Heinemann, 1985.
2. Jain, R. *Planning a Small Scale Industry: A Guide to Entrepreneurs*. S.S. Books, 1984.
3. Kumar, S. A. *Entrepreneurship in Small Industry*. Discovery, 1990.
4. Thomas, Antony. *Information Technology for Office*. Pratibha Publications, n.d.
5. Courter, Gini, and Annette Marquis. *MS-Office 2007*. BPB Publications, 2007.

SUGGESTED READINGS

1. Cliffton, D. S., and D. E. Fylie. *Project Feasibility Analysis*. John Wiley, 1977.
2. Desai, A. N. *Entrepreneur and Environment*. Ashish Publishing House, 1990.
3. McClelland, David C., and W. G. Winter. *Motivating Economic Achievement*. Free Press, 1969.



Mahatma Gandhi University Kottayam

Faculty/ Discipline*	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc. (Honours) Industrial Instrumentation and Automation				
Course Name*	Environmental studies for Instrumentation				
Type of Course*	VAC				
Course Code	MG4VACIIM200				
Course Level*	200				
Course Summary	This course provides a basic understanding of the environment, natural resources, ecosystems, biodiversity, and pollution. It emphasizes sustainable development, conservation, pollution control, and the role of individuals and laws in environmental protection.				
Semester*	4	Credits*		3	Total Hours*
Course Details	Learning Approach	Lecture	Practical *	OJT*	
		3	0	0	
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains *	PO No
1	Understand the basic concepts of environmental studies, types of natural resources, and the principles of sustainable development.	U, An	1,2,3,4,6,7,8,9,10
2	Explain the structure and functioning of ecosystems, recognize various levels of biodiversity, and evaluate the need and methods for biodiversity conservation.	U, An	1,2,3,4,5,6,7,8,9,10
3	Identify various types of pollution, their causes and effects, and evaluate control measures, legal frameworks, and the role of individuals in environmental protection.	U, An	1,2,3,4,5,6,7,8,9,10
4	Students will be able to understand the structure and functioning of ecosystems, identify different levels of biodiversity, and evaluate the importance and methods of biodiversity conservation.	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	2	1	0	3	3	1	1	2
CO 2	3	3	2	1	1	3	3	2	2	2
CO 3	3	3	2	1	1	3	3	3	2	3
CO 4	3	3	2	1	1	3	3	3	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1				
	1.1	Definition and Meaning of Environmental Studies Scope of Environmental Studies (Interdisciplinary Nature) Importance of Environmental Studies in real life, education, policy, and sustainability	3	1
	1.2	Biotic Components: Plants, Animals, Microorganisms, their ecological roles, Abiotic Components: Air, Water, Soil, Light, Temperature, etc. Interaction between Biotic and Abiotic components (e.g., food chains, nutrient cycling)	5	1
	1.3	Renewable Resources and Non-renewable Resources Resource Depletion: Causes, human impact, overuse Conservation Methods: Reduce, Reuse, Recycle, Afforestation, Sustainable use	6	1
	1.4	Concept of Sustainable Development, meaning, goals, Desired outcome, unsustainable to sustainable	4	1
2	2.1	Concept and definition of an ecosystem Structure: Biotic and abiotic components Functions: Energy flow, nutrient cycles (carbon, nitrogen), ecological succession	4	2

	2.2	Concept and definition of an ecosystem Structure: Biotic and abiotic components Functions: Energy flow, nutrient cycles (carbon, nitrogen), ecological succession	5	2	
	2.3	Food chains and food webs (grazing and detritus chains) Ecological pyramids: numbers, biomass, energy Trophic levels and energy transfer efficiency	3		
	2.4	Levels of biodiversity: genetic, species, and ecosystem diversity Importance of biodiversity Threats: habitat loss, climate change, pollution, invasive species Conservation strategies: (In-situ and Ex-situ)	3		
3					
	3.1	Types of Pollution: (Air, Water, Soil, Noise, Thermal And Nuclear) Causes and Effects: Sources, pollutants, impacts on health, environment, and economy Control Measures: Prevention, treatment technologies, regulations, eco-friendly alternatives	5	3	
	3.2	Solid and liquid waste classification Sources of waste: domestic, commercial, industrial Waste disposal and treatment: segregation, recycling, composting, incineration Sustainable waste management practices	4	4	

	3.3	Environmental Laws and Institutions: Environment Protection Act (EPA), 1986 Central Pollution Control Board (CPCB) State Pollution Control Boards (SPCBs)	3	4
4		(Teacher specific content)		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • se of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT
	Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Theory(25 Marks) 1. Written Test 2. Assignment 3. Seminar
	B. End Semester Evaluation (ESE) ESE for Theory : Written Test(50 Marks,1.5 hrs) Part A : Very Short Answer Questions(10 out Of 12 Questions) – (10*2=20 marks) Part B : Short Answer Questions(5 out of 7 Questions)(5*4=20Markss) Part C : Essay Questions(1 out of 2 Questions)-(1*10=10 Marks)

REFERENCES :

1. Erach Bharucha – *Textbook of Environmental Studies for Undergraduate Courses*
2. R. Rajagopalan – *Environmental Studies: From Crisis to Cure*
3. Kaushik, C.P. & Kaushik, Anubha – *Perspectives in Environmental Studies*

SUGGESTED READINGS:

1. Sharma, P. D. – *Ecology and Environment*
2. <https://moef.gov.in/>
3. <https://cpcb.nic.in/>

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Internship				
Type of Course	INT				
Course Code	MG4INTIIM200				
Course Summary	The internship is designed to provide students with real-world exposure and hands-on experience in professional environments aligned with their skill domain and major area of study. It acts as a vital link between academic learning and industry application, allowing students to apply theoretical concepts to practical situations. Through active engagement in industry, research institutions, or academic labs, students gain insights into organizational operations, workplace practices, and professional expectations. The internship also supports the development of key professional competencies such as communication, teamwork, time management, and ethical responsibility. Additionally, it encourages critical thinking, reflection, and self-assessment, helping students identify personal strengths and explore potential career pathways. Students shall undergo the internship in a Firm, Industry, or Organization, or engage in Training in Labs with faculty and researchers, or other Higher Education or Research Institutions, ensuring alignment with their area of academic specialization.				
Semester	4	Duration	60 hours	Credits	2

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
1	Demonstrate practical understanding of operational aspects in Their domain by engaging in real-world industry settings.	Ap	1,3,6,10

2	Apply academic knowledge and skills to identify and solve industry-relevant problems.	A	1,2,3,10
3	Exhibit professional competencies including effective communication, teamwork, time management, and ethical Responsibility.	S	4,5,8,9
4	Develop an understanding of workplace practices, expectations, and challenges.	U	1,6,10
5	Reflect critically on their internship experience to identify Personal strengths, growth areas, and career aspirations.	E	1,6,10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

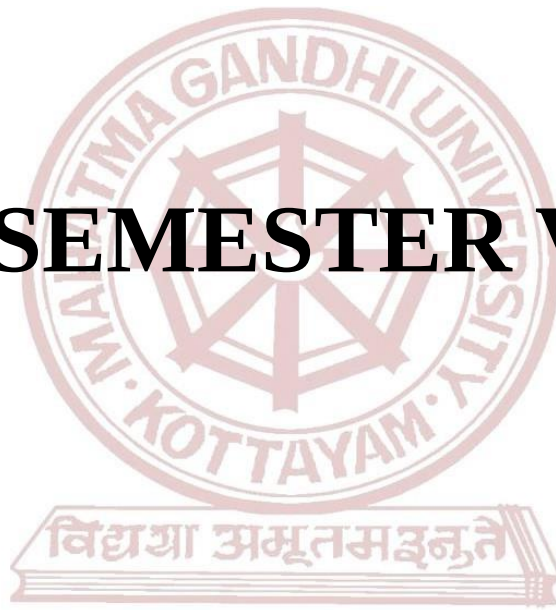
CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	0	3	0	0	2	0	0	0	2
CO 2	3	2	3	0	0	0	0	0	0	2
CO 3	0	0	0	3	2	0	0	2	2	0
CO 4	3	0	0	0	0	2	0	0	0	2
CO 5	3	0	0	0	0	2	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	MODE OF ASSESSMENT		
	A	Continuous Comprehensive Assessment (CCA)	
		Components	Marks
		Feedback from the hosting organization	5
		Internal Supervisor feedback	10
		Total	15
	B	End Semester Evaluation (ESE)	
		Components	Marks
		Presentation	10
		Report	10
		Viva Voce	15
		Total	35

SEMESTER V



MGU-B.VOC. (HONOURS)

Syllabus

Semester: 5

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG5SDCIIM300	Industrial Communication Protocols	SDC	4	4	4	0	0
MG5SDCIIM301	AI based Robotic Automation	SDC	4	5	3	2	0
MG5SDEIIM300	Process Control Instrumentation	SDE (Any One)	4	5	3	2	0
MG5SDEIIM301	Industry Layouts						
MG5SDEIIM302	Power Engineering						
MG5SECIIM300	Virtual Instrumentation	SEC	3	4	2	2	0
MG5MPCIIM300	Process Control Communication	MPC	4	4	4	0	0
MG5VACIIM300	Digital Empowerment	VAC	3	3	3	0	0



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Industrial Communication Protocols			
Type of Course	SDC			
Course Code	MG5SDCIIM300			
Course Level	300			
Course Summary	The Industrial Data Communication (IDCN) course is designed to provide students with a comprehensive understanding of the communication systems and protocols used in industrial automation and control systems. This course is typically offered to students in Instrumentation & Control Engineering (ICE) and Electronics & Instrumentation Engineering (EIE) disciplines, often in the 7th or 8th semester of their undergraduate studies.			
Semester	5	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		4	0	0
Pre- requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the classification, characteristics, and selection of transducers and sensors used in measurement systems.	U	1,2,3, 4,6,7, 8,9,10
2	Understand the basics of protocols and their necessity in network communication.	U	1,2,3, 4,6,7, 8,9,10
3	Identify the different types of transmission media, including unguided and guided media, wired and wireless, UTP, coaxial, and fiber optic cables.	A	1,2,3, 4,6,7, 8,9,10
4	Understand the Profibus standard, protocol stack, physical layer, data link layer, application layer, and troubleshooting techniques.	U	1,2,3, 4,5,6, 7,8,9, 10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	2	3	2	1	0	1	1	1	2	2
CO 3	2	3	2	1	0	1	1	1	2	2
CO 4	3	3	3	2	1	2	1	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Local Area Network			1
	1.1	Computer networks in instrumentation, components of computer networks- hardware and software, network topologies star, ring, bus, mesh	5	
	1.2	Network classification based on transmission technologies- point to point, broadcast	5	
	1.3	LAN, WAN, MAN, VPN, internet. based on architecture-peer-to-peer, client server, advantages of client server over peer-to-peer model	5	
2	Network devices and communication protocol			2
	2.1	basics of protocol and its need, brief functional description of each the OSI- ISO reference model layers with list of protocols	5	
	2.2	The TCP / IP reference model:brief functional description of each of the layer with list of protocols IP layer protocols ipv4 and IPv6 frame formats internet addressing, network addressing, subnets and subnet masking, gateway dressing	5	
	2.3	broadcast addressing, dotted decimal notation, loopback addressing. Domain name system DNS introduction, mapping to IP address	5	

	Network media and hardware			
3	3.1	Transmission Media: unguided and guided media, wired and wireless, UTP, coaxial and fibre optical cable. types of Connectors: RJ-45, RJ-11, BNC, BNC-T, BNC Terminator,	5	3
	3.2	Fiber optic connectors:- Subscriber Channel(SC), Straight Tip(ST), Mechanical transfer-registered jack(MT-RJ) connectors Network Interface Card (NIC), ARCNET,	5	
	3.3	Ethernet. Network connecting devices: Repeater, Hub, Bridge, Switch, Router, Gateway, Access point, Wireless Access points. Servers introduction : File, Print, Mail, Proxy, Web	5	
		Basics of fieldbus and profibus		
4	4.1	Introduction to Foundation fieldbus physical layer and wiring rules, data link layer, application layer, user layer.	5	4
	4.2	Wiring and installation practice with fieldbus termination and preparation, installation of the complete system, ,	5	
	4.3	Test equipment for foundation fieldbus. Introduction to profibus standard profibus protocol stack physical layer data link layer application layer troubleshooting of profibus.	5	
5		Teacher's specific content		

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Theory(30 Marks) 1. Written Test 2. Seminars 3. Assignments
	B. End Semester Evaluation (ESE) ESE for Theory : Written Test(70 Marks , 2 hrs) Part A: Very Short Answer Questions (Answer all)- (10*2=20 Marks) Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C:Essay Questions(2 out of 3 Questions) - (2*15=30 Marks)

Syllabus

REFERENCES:

1. Tanenbaum, Andrew S., and David J. Wetherall. *Computer Networks*. 5th ed., Pearson Education, New Delhi, 2011. Print.
2. Stallings, WillliIM. *Data and Computer Communications*. Latest ed., PHI Learning, New Delhi. Print.

SUGGESTED READING:

1. Trivedi Bhushan, *Computer networks*, Oxford University Press, New Delhi 2013
2. Forouzen, *Data communication and networking*, - Tata McGraw Hill, Education New Delhi (Latest edition)



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	AI based Robotic Automation				
Type of Course	SDC				
Course Code	MG5SDCIIM301				
Course Level	300				
Course Summary	To introduce the functional elements of Robotics. This course is designed to introduce and familiarize students with programming using the Arduino Micro-controller and controlling robots using Arduino.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basics of robotics, including types, components, and key concepts.	U	1,2,3,4,6,7,8,9,10
2	Understand the construction, types, and control of manipulators and grippers used in robotic systems.	U	1,2,3,4,5,6,7,8,9,10
3	Understand basic Arduino programming and sensor interfacing for building simple robotic automation systems.	An	1,2,3,4,5,6,7,8,9,10
4	Develop hands-on skills in basic Arduino programming and interfacing through simple experiments	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	3	3	2	1	1	1	1	1	2	2
CO 3	3	3	3	1	1	1	1	1	2	3
CO 4	3	3	3	2	2	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT MGU-B.VOC. (HONOURS)

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Robotics- Basic concepts			
	1.1	Definition of robotics Origin of robotics Different types of robots	2	1
	1.2	Degree of freedom Asimov's laws of robotics Dynamic stabilization of robots	4	1

1	1.3	Types of motors Motor controllers Control motor speed Types of sensors	4	1
	1.4	Nickel cadmium batteries Path determination Vision Ranging Laser,	5	1
2	Manipulators, actuators and grippers			
	2.1	Construction of manipulators Manipulator dynamics force control.	5	2
	2.2	Electronics manipulator pneumatic manipulator control circuits	3	2
	2.3	pneumatic hydraulic electric actuators effectors	3	2
	2.4	various types of grippers design considerations.	4	2
	Arduino Programming concept			
3	3.1	Arduino Logic Building Robotic Automation Basic programming Basic of Relay	2	3
	3.2	Arduino for Building Robotics Applications, Advantages of using Arduino	5	3
	3.3	Concepts of Different types of sensors Application of different sensors	4	3
	3.4	Interfacing with single sensor Interfacing with multiple sensors	4	3
4	Practical session			
	Lab experiments using Arduino			

	4.1	1. Familiarization of Arduino 2. Blinking LED 3. Button Controlled LED 4. Traffic Light System	10	4
	4.2	5. LED Fading with PWM 6. Buzzer Sound Control 7. Temperature Monitoring with LM35 Sensor 8. Light Detection using LDR 9. Ultrasonic Distance Measurement 10. Servo Motor Control 11. Potentiometer Controlled LED Brightness. 12. LCD Display with Arduino	20	4
5	Teacher Specific Contents			



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Process control instrumentation				
Type of Course	SDE				
Course Code	MG5SDEIIM300				
Course Level	300				
Course Summary	This course provides a foundational understanding of control theory and process control instrumentation, covering system modeling, controller design, control elements, and practical implementation techniques used in industrial automation.				
Semester	5	Credits			Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	Syllabus				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Students will understand the basic structure and behavior of open and closed loop control systems, and apply Laplace transform techniques to analyze control system models.	U	1,2,3,4,6,7,8,9,10
2	Students will be able to explain and differentiate various controller types and actions, and evaluate the suitability of control strategies for different industrial processes.	U	1,2,3,4,5,6,7,8,9,10
3	Students will identify and describe the function, types, and selection criteria of final control elements such as control valves, actuators, and signal converters in process control	An	1,2,3,4,6,7,8,9,10
4	Students will gain hands-on experience in implementing and tuning control systems using simulators and hardware to observe system responses and controller effects.	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	3
CO 2	3	3	2	1	1	1	1	1	2	3
CO 3	2	3	2	1	0	1	1	1	2	2
CO 4	3	3	3	2	1	1	1	1	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Syllabus

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Fundamentals of Control Systems			
	1.1	Basics of open loop and closed loop systems	2	1
	1.2	Introduction to Laplace transforms and inverse Laplace transforms	4	1
	1.3	Block diagram reduction techniques	4	1
	1.4	Signal flow graphs and Mason's Gain Formula Practical examples of control loops	3	1
2	Controllers and Control Actions			
	2.1	Types of controllers: ON-OFF, Proportional, Integral, Derivatives	5	2
	2.2	Introduction to PI, PD, and PID control Controller modes and tuning: Proportional Band, Offset.	5	2
	2.3	Selection criteria for controllers Overview of electronic and pneumatic controllers	5	2
	Control Valves and Final Control Elements			
	3.1	Actuators: Electric, Pneumatic, Hydraulic	2	3

Syllabus

3	3.2	Control valve types: Linear, Equal percentage, Quick opening	5	3
	3.3	Valve components: Solenoid valve, Split range, Seat plugs Concepts: Valve sizing, Cv rating, Cavitation and Flashing effects	4	3
	3.4	Introduction to signal converters (P/I and I/P)	4	3
4	(Practical Session)			
	Lab Experiments			
	4.1	1. Study of open loop and closed loop systems using a simulator 2. Tuning of a PID controller using Ziegler-Nichols method 3. Simulation of a level control system using software (e.g., MATLAB/Simulink or SCILAB) 4. Demonstration of flapper-nozzle pneumatic system	15	4
	4.2	1. Measuring response of flow control system with transportation lag 2. Observation of control valve characteristics (equal %, linear, quick opening) 3. I/P and P/I converter working demonstration 4. Basic programming of temperature control using Arduino or PLC	15	4
5	Teacher Specific Contents			



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation					
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation					
Course Name	Industry Layouts					
Type of Course	SDE					
Course Code	MG5SDEIIM301					
Course Level	400					
Course Summary	This course introduces students to Piping and Instrumentation Diagrams (P&ID), which are essential tools used in the design, operation, and maintenance of industrial systems. The focus of the course is to understand the symbols, standards, and practices involved in creating and interpreting P&IDs for various processes, particularly in industries like chemical, oil & gas, power generation, and manufacturing.					
Semester	5	Credits			4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT		
		3	1	0	75	
Pre- requisites, if any	NIL					

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify standard process line symbols used in Piping and Instrumentation Diagrams	U	1,2,3,4,9,10
2	Interpret flow sheet code letter services and apply correct codes for process and utility systems.	U	1,2,3,4,9,10
3	Identify functional designations for devices using standard symbols and alphabetic coding (A–Z).	U	1,2,3,4,9,10
4	Identify transmission and switching symbols, including timers, relays, and interlocks used in Programmable Logic Controllers (PLCs).	U	1,2,3,4,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			



MGU-B.VOC. (HONOURS)

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	2	1	0	0	0	0	1	1
CO 2	3	3	2	2	0	0	0	0	1	1
CO 3	3	2	2	2	0	0	0	0	1	1
CO 4	3	2	2	2	0	0	0	0	1	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Process equipment symbols			1
	1.1	Process line symbols	5	
	1.2	Symbols of valves, Symbols of actuator with and without positioner	5	
	1.3	Furnace and boiler symbols, Heat transfer symbols ,Pump and compressor, Process pressure vessels, Dryers	5	
2	Process equipment			2
	2.1	Flow sheet code letter service, Symbol for Lines Utility symbols Drainer, strainer, filter, mixer, twin basket filter, spray nozzle	5	
	2.2	steam separator, bootleg, seal legs, , gage hatch, man-hole, flame arrester,	5	
	2.3	Symbols of function designations for relays, Symbols for primary location, field mounted, auxiliary location for discrete instruments, shared display and control	5	
3	Instrument identification			3
	3.1	Instrument identification or tag number, identification of letter Instrument line symbols	5	
	3.2	PLC Symbols for self-actuated regulators, valves and other devices for: Flow, Level, Pressure, Temperature, Traps, Multiple way valves,	5	
	3.3	Primary element symbols (A to Z), Function symbols (A to Z)	5	

4	Practical Session			4
	4.1	1. Draw the symbols of Logical function symbols Gates, Transmission and Switching symbols, Timer, Switching symbols for PLC- Push buttons, Foot switch,	10	
	4.2	2. Draw the symbols of Limit switch, Process parameter switch, Output device, Distributed control / shared display symbols (As per ISA	10	
	4.3	3. Draw the symbols of Distributed control logic/sequential Control Symbols- Normally accessible to operator, Not normally accessible to operator. Miscellaneous symbols- Computing/signal conditioning, software system link 4. Identify and paste symbols of equipments	10	
5	Teacher Specific Contents			

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Power Engineering				
Type of Course	SDE				
Course Code	MG5SDEIIM302				
Course Level	300				
Course Summary	Helps students understand basic electronics, semiconductor devices, and digital systems, and develop skills to identify components and design simple electronic circuits.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the fundamental concepts and components used in power electronics systems.	U	1,2,3,4, 6,7,8,9,10
2	Analyze the operation and characteristics of various rectifiers and converters.	U	1,2,3,4, 6,7,8,9,10
3	Demonstrate knowledge of inverter and chopper circuits and their practical applications.	An	1,2,3,4, 5,6,7,8,9, 10
4	Apply power electronic controllers in industrial and domestic systems effectively.	A	1,2,3,4, 5,6,7,8,9, 10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	3	3	2	1	0	1	1	1	1	2
CO 3	3	3	2	1	1	1	1	1	2	3
CO 4	3	3	3	2	2	2	2	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	INTRODUCTION TO POWER ELECTRONICS			
	1.1	Basics of Power Electronics Characteristics and Ratings of Devices	2	1

1	1.2	Applications and Advantages	4	1
	1.3	Power Semiconductor Devices (Diodes, SCRs, MOSFETs, IGBTs)	4	1
	1.4	Characteristics and Ratings of Devices	5	1
2	RECTIFIERS AND CONVERTERS			
	2.1	Uncontrolled Rectifiers (Half and Full Wave) PN Junction Diode: structure, working, VI characteristics.	5	2
	2.2	AC to DC Converters	5	2
	2.3	Output Voltage Control Techniques	5	2
	INVERTERS AND CHOPPERS			
3	3.1	DC to AC Converters (Inverters)	2	3
	3.2	Classification and Operation of Inverters	5	3
	3.3	Choppers – Step-up and Step-down	4	3
	3.4	Applications of Choppers and Inverters	4	3
4	(Practical Session)			
	Lab Experiments			
	4.1	1. Study and testing of Power Diode and SCR 2. VI Characteristics of SCR 3. VI Characteristics of MOSFET and IGBT	10	4

	4.2	4. Half-Wave Rectifier using Diode and SCR 5. Full-Wave Controlled Rectifier using SCRs 6. Single Phase AC Voltage Controller using TRIAC 7. Step-Down Chopper using MOSFET 8. Step-Up Chopper using IGBT 9. Single Phase Inverter using MOSFET 10. Speed Control of DC Motor using Chopper	20	4
5	Teacher Specific Contents			



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Virtual Instrumentation				
Type of Course	SEC				
Course Code	MG5SECIIM300				
Course Level	300				
Course Summary	Students will be able to graphically represent and simulate data acquisition processes using virtual instrumentation tools.				
Semester	5	Credits			3
Course Details	Learning Approach	Lecture	Practical	OJT	Total Hours
		2	1	0	
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Students will understand the fundamentals of virtual instrumentation and basic programming techniques used in virtual instruments.	U	1,2,3,4,5,6,7,8,9,10
2	Students will learn the basics of data acquisition systems and how to interface them with a PC using analog and digital inputs/outputs.	U	1,2,3,4,5,6,7,8,9,10
3	Knowledge of data acquisition, signal processing and analysis techniques.	An	1,2,3,4,5,6,7,8,9,10
4	Students will gain the ability to create simple virtual instruments using LabVIEW and apply basic dataflow programming concepts	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	2	1	1	1	2	2	3
CO 2	3	3	2	2	1	1	1	2	2	3
CO 3	2	3	2	2	1	1	1	1	2	2
CO 4	3	3	2	2	2	1	1	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

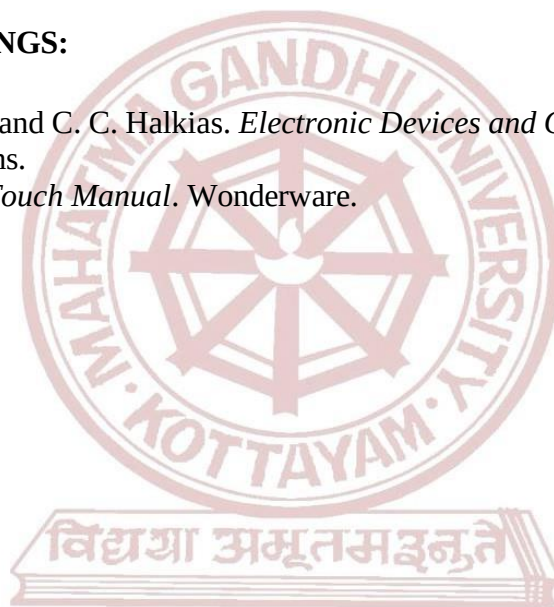
Module	Units	Course description	Hrs	CO No.
1	Virtual Instrumentation Basics			
	1.1	-Definition and Advantages over Traditional Instrument	4	1
	1.2	- Basic VI Programming Techniques	4	1
	1.3	- Loops and Charts	4	1
	1.4	- Arrays and Graphs - Simple File Input/Output	4	1
2	- Basics of Data Acquisition (DAQ)			
	2.1	- Understanding Analog Input and Output	2	2
	2.2	PC-Based DAQ System Overview	4	2
	2.3	- Digital Input and Output - Real-Time System Integration Basics - Introduction to LabVIEW	4	2
	2.4	- Creating a Simple Virtual Instrument - Using Arrays and File Input/Output in LabVIEW	4	2
	-(Practical Session)			
3	3.1	1.Basic Arithmetic Operations 2. Odd or Even	10	3,4

REFERENCES:

1. Gupta, S., and J. P. Gupta. *PC Interfacing for Data Acquisition and Process Control*. Instrument Society of America, 1994.
2. Gofton, Peter W. *Understanding Serial Communications*. Sybex International, latest ed.
3. Bishop, Robert H. *Learning with LabVIEW*. Prentice Hall, 2003.
4. James, Kevin. *PC Interfacing and Data Acquisition: Techniques for Measurement, Instrumentation and Control*. Newnes, 2000.

SUGGESTED READINGS:

1. Millman, Jacob, and C. C. Halkias. *Electronic Devices and Circuits*. TMH Publications.
2. *Wonderware InTouch Manual*. Wonderware.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Subject	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Process Control Communication			
Type of Course	MPC			
Course Code	MG5MPCIIIM300			
Course Level	300			
Course Summary	The Industrial Data Communication (IDCN) course is designed to provide students with a comprehensive understanding of the communication systems and protocols used in industrial automation and control systems. This course is typically offered to students in Instrumentation & Control Engineering (ICE) and Electronics & Instrumentation Engineering (EIE) disciplines, often in the 7th or 8th semester of their undergraduate studies			
Semester	5	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		4	0	60
Pre- requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the classification, characteristics, and selection of transducers and sensors used in measurement systems.	U	1,2,3, 4,6,7, 8,910
2	Understand the basics of protocols and their necessity in network communication.	U	1,2,3, 4,5,6, 7,8,9 10
3	Identify the different types of transmission media, including unguided and guided media, wired and wireless, UTP, coaxial, and fiber optic cables.	A	1,2,3, 4,6,7, 8,910
4	Understand the Profibus standard, protocol stack, physical layer, data link layer, application layer, and troubleshooting techniques.	U	1,2,3, 4,5,6, 7,8,9 10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	2	1	0	1	1	1	1	2
CO 2	2	2	3	2	1	1	1	1	2	2
CO 3	2	3	2	1	0	1	1	1	1	2
CO 4	3	3	3	2	1	1	1	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Local Area Network			
	1.1	Computer networks in instrumentation, components of computer networks- hardware and software, network topologies star, ring, bus, mesh	5	1
	1.2	Network classification based on transmission technologies- point to point, broadcast	5	
	1.3	LAN, WAN, MAN, VPN	5	
2	Network devices and communication protocol			
	2.1	basics of protocol and its need, brief functional description of each the OSI- ISO reference model layers with list of protocols	5	2
	2.2	The TCP / IP reference model:brief functional description of each of the layer with list of protocols	5	
	2.3	IP layer protocols ipv4 and IPv6 frame formats	5	
	Network media and hardware			
3	3.1	Transmission Media:unguided and guided media, wired and wireless, UTP, coaxial and fibre optical cable. types of	5	3

		Connectors: RJ-45, RJ-11, BNC, BNC-T, BNC Terminator,		
	3.2	Fiber optic connectors:- Subscriber Channel(SC), Straight Tip(ST), Mechanical transfer-registered jack(MT-RJ) connectors Network Interface Card (NIC)	5	
	3.3	Ethernet. Network connecting devices: Repeater, Hub, Bridge, Switch, Router, Gateway, Access point, Wireless Access points. Servers	5	
		Basics of fieldbus and profibus		
	4.1	Introduction to Foundation fieldbus physical layer and wiring rules, data link layer, application layer, user layer.	7	
	4.2	Introduction to profibus standard profibus protocol stack physical layer data link layer application layer troubleshooting of profibus	8	
4				4
5		Teacher's specific content		



MGU-B.VOC. (HONOURS)

Syllabus

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT Mode of Assessment
	A. Continuous Comprehensive Assessment (CCA) CCA for Theory(30 Marks) 1.Written Test 2.Seminars 3.Assignments
	B. End Semester Evaluation (ESE) ESE for Theory : Written Test(70 Marks , 2 hrs) Part A: Very Short Answer Questions (Answer all)- (10*2=20 Marks) Part B: Short Answer Questions (4 out of 6 Questions) - (4*5=20 Marks) Part C:Essay Questions(2 out of 3 Questions) - (2*15=30 Marks)

REFERENCES:

1. Tannebaum AndrewS Wetherall David J., *Computer networks* Pearson, New Delhi 5th Edition 2011
2. Stallings WilliIMs, *Data and computer communications*, PHI Learning, New Delhi, Last Edition

SUGGESTED READING:

1. Trivedi Bhushan, *Computer networks*, Oxford University Press, New Delhi 2013
2. Forouzen, *Data communication and networking*, - Tata McGraw Hill, Education New Delhi (Latest edition)

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University

Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Digital Empowerment				
Type of Course	VAC				
Course Code	MG5VACIIM300				
Course Level	300				
Course Summary	This course, "Digital Empowerment," provides an engaging exploration of fundamental digital life manifested in everyday life, such as various e services provided by the Governments. Also provide a hands own experience, in creating accounts and handling them form the real world scenario. Also enable students to handle basic document, spreadsheet and presentation preparations.				
Semester	5	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	0	0	
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	To familiarises various day to day digital applications provided by Governments	U	1,2,3,4,5, 6,7,8,9,10
2	To get hands own training in creation and handling various digital applications	U	1,2,3,4,5, 6,7,8,9,10
3	To get a basic knowledge in word processor and hands on training for the preparation of a document	A	1,2,3,4,5, 6,7,8,9,10
4	To get basic knowledge in spread sheets and hands on experience in handling a spread sheet	An	1,2,3,4,5, 6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	2	2	2	2	2	2	2	3
CO 2	3	2	2	1	2	2	2	2	2	2
CO 3	2	2	2	2	2	2	2	2	2	3
CQ4	3	3	3	2	1	3	3	3	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
	Digital inclusion			
1	1.1	Needs and challenges, Vision of Digital India: DigiLocker,	4	1
	1.2	e-Pathshala, BHIM,	5	1
	1.3	Public utility portals of Govt. of India such as RTI, Health, Finance, Income Tax filing, Education	4	1
	1.4	Electronic Communication: electronic mail, blogs, social media • Collaborative Digital platforms • Tools/platforms for online learning • Collaboration using file sharing, messaging, video conferencing	6	1
	Daily Life applications			
2	2.1	Create a Digi locker account and save certificates in the Digi locker	4	2
	2.2	Demonstrate how to create a e-sanjeevani or UHID account and how to take the appointment of a doctor online	4	2
	2.3	Demonstrate how income tax filing can be done for a person	4	2
	2.4	Demonstrate how to register in SWAYAM portal and how to register a course	4	2

	Digital Empowerment			
3	3.1	E-Hospitals	4	3
	3.2	e-Kranti (Electronic Delivery of Services)	3	3
	3.3	Collaborative Digital platforms .Tools/platforms for online learning	3	4
4	Teacher Specific Contents			

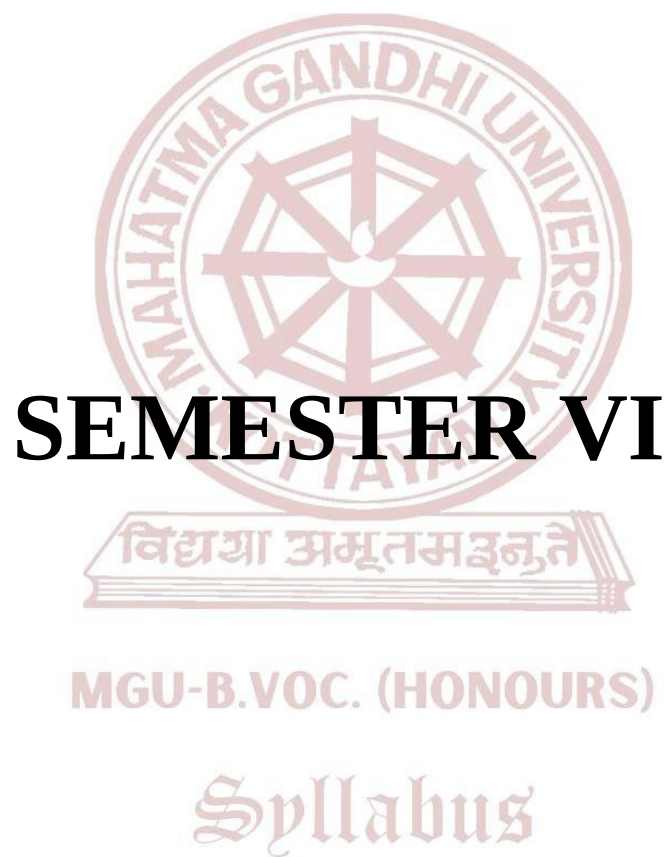
Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) CCA for Theory(25 Marks) 1. Written Test 2. Assignments 3. Seminars
	C. End Semester Evaluation (ESE) ESE for Theory : Written Test (50 Marks, 1.5 hrs) Part A : Very Short Answer Questions(10 out of 12 Questions)-(10*2=20 Marks) Part B : Short Answer Question (5 out of 7 Questions) – (5*4=20Marks) Part C : Essay Questions(1 out of 2 Questions) – (1*10=10 Marks)

REFERENCES:

1. Rodney Jones and Christoph Hafner. "Understanding digital literacies: A practical Introduction". Routledge Books, 2nd edition, 2021. 12
2. Suggested Readings • David Sutton. "Cyber security: A practitioner's guide", BCS Learning & Development Limited, UK, 2017.

SUGGESTED READINGS:

1. *Digital India: Unlocking the Trillion Dollar Opportunity* – Pradip N. Khandwalla
2. *Digital Empowerment: Access to Information and Communication Technology* – Richard Heeks



SEMESTER VI

MGU-B.VOC. (HONOURS)

Syllabus

Semester: 6

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG6SDCIIM300	Applications of Programmable Logic Controller (PLC) and Supervisory Control and Data Acquisition (SCADA) in Industry	SDC	4	5	3	2	0
MG6SDEIIM300	Advanced RISC Machine (ARM) and VEGA Processors	SDE (Any One)	4	5	3	2	0
MG6SDEIIM301	Communication Electronics						
MG6SECIIM300	Solar Technology	SEC	3	3	3	0	0
MG6MPCIIM300	Automation Controllers	MPC	4	4	4	0	0
MG6VACIIM300	Industrial Safety Measures	VAC	3	3	3	0	0
MG6PRJIIM300	Project	PRJ	4	8	0	8	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Applications of Programmable Logic Controller(PLC)and Supervisory Control and Data Acquisition(SCADA) in Industry				
Type of Course	SDC				
Course Code	MG6SDCIIM300				
Course Level	300				
Course Summary	This course provides a comprehensive introduction to Programmable Logic Controllers (PLCs) and Supervisory Control and Data Acquisition (SCADA) systems, which are fundamental technologies in industrial automation and process control. Students will learn the history, architecture, and operation principles of PLCs, including the various hardware components such as CPUs, discrete and analog I/O modules, memory types, and programming devices. The course covers practical aspects like wiring standards, programming techniques, and troubleshooting PLC systems.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	
Pre-requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify and explain the various parts of a PLC including the CPU, memory units, power supply, and I/O modules.	U	1,2,3,4,10
2	Apply Boolean algebra rules to simplify logic expressions and analyze logic circuits.	U,A	1,2,3,4,10
3	Identify different types of input devices such as push button switches, toggle switches, proximity switches, and photoelectric switches.	U	1,2,3,4,10
4	Identify and differentiate between various PLC programming languages with emphasis on ladder logic.	U	1,2,3,4,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	1	1	0	0	0	0	0	2
CO 2	3	3	1	1	0	0	0	0	0	2
CO 3	2	2	1	1	0	0	0	0	0	2
CO 4	3	3	2	1	0	0	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT MGU-B.VOC. (HONOURS)

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	PLC Programmable Logic Controllers (PLCs):			
	1.1	Controller modes, definition & history of the PLC; Principles of Operation; Various Parts of a PLC	4	1

	1.2	The I/O section, Discrete I/O Modules, Analog I/O Modules, Special I/O Modules, I/O specifications, UAVs for crop monitoring, spraying, and data collection.	4	1
	1.3	Types, Programming Devices, Selection of wire types.	2	1
	1.4	Introduction; The CPU, Memory design, Memory	5	1
	Fundamentals of Logic			
2	2.1	The Binary Concept, AND, OR and NOT functions, Boolean Algebra, developing circuits from Boolean Expression expressions,	5	2
	2.2	Producing the Boolean equation from given circuit, Hardwired logic versus programmed logic, Programming word level logic instructions	3	2
	2.3	Converting Relay schematics and Boolean equation into PLC Ladder Programs,	3	2
	2.4	Writing a ladder logic program directly from a narrative description	4	2
	Various INPUT /OUTPUT Devices and its interfacing with PLC.			
3	3.1	Different types of Input devices: Switches: Push button Switches, Toggle Switches, Proximity switches, Photo switches,	2	3

	3.2	Temperature Switch, Pressure Switch, and Level Switch, Flow Switches	5	3
	3.3	Transducers and sensors, , manually operated switches, Motor starters	4	3
	3.4	Transmitters Their working, specification and interfacing	4	3
4	(Practical Session)			
	Lab Experiments			
	4.1	1. Device a program to add, subtract in ladder diagram 2. Device a program to multiply, divide in ladder diagram 3. Familiarize various counters	15	4
	4.2	4. Familiarize various timers 5. Motor 6. basic logic gates 7. Half adder, full adder	15	4
5	MGU-B.VOC. (HONOURS) Teacher Specific Contents Syllabus			



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Advanced RISC Machine (ARM) and VEGA Processors				
Type of Course	SDE				
Course Code	MG6SDEIIM300				
Course Level	300				
Course Summary	This course provides a foundational understanding of ARM and VEGA processors, covering their architecture, instruction sets, and programming models. Students will explore various ARM instruction types and gain hands-on experience by writing and executing basic ARM assembly programs. The course also introduces the RISC-V-based VEGA processors, highlighting their features, architecture, and role in AI-enabled embedded applications.				
Semester	6	Credits			Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the basics of ARM processors, including their architecture, programming, and instruction types.	U	1,2,3,4,6,7,8,9,10
2	Learn different types of ARM instructions, including data processing, data transfer, control flow, and co-processor instructions.	U	1,2,3,4,6,7,8,9,10
3	Understand the basics of VEGA processors, their architecture.	U	1,2,3,4,6,7,8,9,10
4	Learn to write and execute basic ARM assembly language programs for arithmetic operations and logical problem solving.	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	1	1	1	1	2
CO 2	3	3	2	1	0	1	1	1	1	2
CO 3	2	2	2	1	0	1	1	1	1	2
CO 4	3	3	2	2	1	1	1	1	2	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	ARM processor			
	1.1	Advanced RISC Machine Architecture Inheritance ARM Programming Model	2	1
	1.2	ARM Development Tools	2	1

	1.3	3 and 5 stages Pipeline ARM Organization.	2	1
	1.4	ARM Instruction Execution and Implementation . ARM Co-Processor Interface.	4	1
	1.5	ARM Instruction Types – Data Transfer Data Processing Control Flow Instructions	5	1
2	ARM Instruction set			
	2.1	ARM Instruction Set	5	2
	2.2	Co-Processor Instructions	3	2
	2.3	Data Processing Instruction Data Transfer Instruction	3	2
	2.4	Control Flow Instructions	4	2
	VEGA Processor			
3	3.1	Introduction to VEGA Processors Overview of VEGA processors – Designed in India (by C-DAC) Importance of RISC-V architecture Features and benefits of VEGA processors	2	3
	3.2	Architecture of VEGA Processor Basic block diagram Instruction set architecture (ISA) Memory organization and I/O structure	5	3

	3.3	AI Capabilities in VEGA Processors Integration of AI/ML accelerators Support for neural network processing Edge AI applications using VEGA	4	3
	3.4	VEGA Processor Development Tools Introduction to VEGA SDK Code compilation and debugging tools Simulation environments	4	3
4	Practical session			
	Lab experiments using Arm Processor using KEIL Software			
	4.1	1. Introduction to Arm 2. Write a program in assembly language to find addition of two numbers 3. Write a program in assembly language to find subtraction of two numbers. 4. Write a program in assembly language to find Multiplication of two numbers.	10	4
	4.2	5. Write a program in assembly language to find Multiplication of two numbers. 6. Write a program in assembly language to find 64 bit Multiplication of two numbers. 7. Write a program in assembly language to find division of two numbers. 8. Write a program in assembly language to find factorial of a number.	20	4
5	Teacher Specific Contents			



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Communication Electronics				
Type of Course	SDE				
Course Code	MG6SDEIIM301				
Course Level	300				
Course Summary	Helps students understand basic electronics, semiconductor devices, and digital systems, and develop skills to identify components and design simple electronic circuits.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	NIL				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Students will understand the basic principles, history, components, and practical applications of radio communication systems.	U	1,2,3,4,5,6,7,8,9,10
2	Understand modern communication technologies	U	1,2,3,4,5,6,7,8,9,10
3	Demonstrate knowledge of inverter and chopper circuits and their practical applications.	An	1,2,3,4,5,6,7,8,9,10
4	Apply power electronic controllers in industrial and domestic systems effectively.	A	1,2,3,4,5,6,7,8,9,10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	2	1	1	1	2	2	3
CO 2	2	3	1	1	1	1	1	2	1	2
CO 3	3	3	2	2	2	1	1	2	3	2
CO 4	3	3	2	2	2	1	1	2	3	3

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Radio Communication			
	1.1	Introduction to Radio Communication. What is Radio Communication?	2	1

1	1.2	Electromagnetic Waves Frequency, wavelength, and propagation	4	1
	1.3	History and evolution Basic Components of a Radio System Transmitter, receiver, antenna	4	1
	1.4	Applications of Radio Communication Broadcasting, walkie-talkies, aviation, etc.	5	1
2	Radio Wave Propagation			
	2.1	Radio Wave Propagation Types of Propagation Ground wave Skywave (ionospheric) Line-of-sight (space wave)	5	2
	2.2	Frequency Bands and Uses HF, VHF, UHF, etc.	5	2
	2.3	Factors Affecting Propagation Weather, terrain, obstacles, ionosphere	5	2
3	Modulation/Demodulation Techniques			
	3.1	Modulation Techniques Need for Modulation	2	3
	3.2	Types of Modulation Amplitude Modulation (AM)	5	3
	3.3	Frequency Modulation (FM) Single Sideband (SSB)	4	3

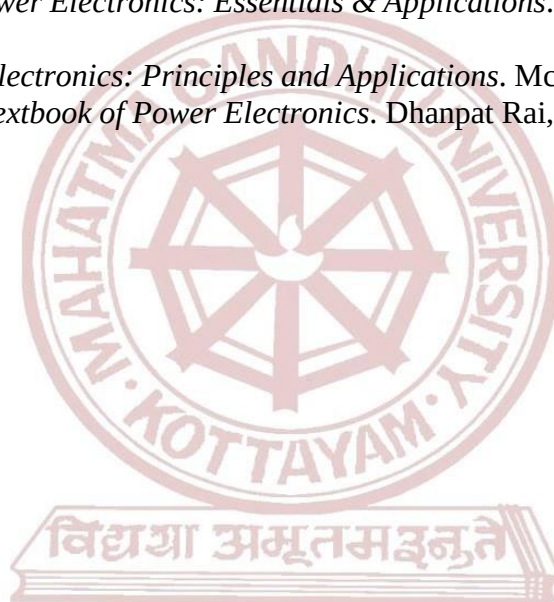
	3.4	Demodulation Basics	4	3
		Extracting the original signal		
4	(Practical Session)			
	Lab Experiments			
	4.1	1. Introduction to Radio Equipment Objective: Identify and understand the basic components of a radio system (transmitter, receiver, antenna). 2. Generation of AM Signal using Function Generator Objective: Create an amplitude modulated signal and observe it on a CRO. 3. Detection of AM Signal using Diode Detector Objective: Extract audio from an AM wave using a simple diode demodulator circuit. <i>Apparatus:</i> AM signal source, diode, capacitor, earphone. 4. Generation of FM Signal using Function Generator Objective: Produce an FM signal and examine frequency variations with modulating input. <i>Apparatus:</i> Function generator with FM option, CRO. 5. Introduction to SDR (Software Defined Radio) Objective: Use an SDR dongle to receive AM/FM signals and display them on a PC.	30	4

REFERENCES

1. Tomasi, W. (2003). *Electronic Communications Systems: Fundamentals through Advanced* (5th ed.). Pearson/Prentice Hall.
2. Taub, H., & Schilling, D. L. (1986). *Principles of Communication Systems* (2nd ed.). McGraw-Hill.
3. Lathi, B. P., & Ding, Z. (2019). *Modern Digital and Analog Communication Systems* (5th ed.). Oxford University Press.
4. Sizun, H. (2006). *Radio Wave Propagation for Telecommunication Applications* (illustrated ed.). Springer Science & Business Media.

SUGGESTED READING:

1. Umanand, L. *Power Electronics: Essentials & Applications*. Wiley India, 2009.
2. Joseph. *Power Electronics: Principles and Applications*. McGraw-Hill, 1995.
3. Singh, S. N. *A Textbook of Power Electronics*. Dhanpat Rai, 2007.



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation and Automation				
Course Name	Solar technology `				
Type of Course	SEC				
Course Code	MG6SECIIM300				
Course Level	300				
Course Summary	This course is designed to meet the growing demand for skilled professionals in the renewable energy sector, specifically in the field of solar photovoltaic.				
Semester	6	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	0	0	45
Pre- requisites, if any	NIL				

MGU-B.VOC. (HONOURS)

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Demonstrate the basics of PV based power plant	U	1,2,3, 4,6,7, 8,9,10
2	Develop a solar power plant based on the estimation of power requirement	A	1,2,3, 4,5,6, 7,8,9, 10
3	Analyse and troubleshoot issues in solar power system for recycling electronic products	U	1,2,3, 4,5,6, 7,8,9, 10
4	Design an expertise in the installation of Solar power plant	U	1,2,3, 4,5,6, 7,8,9, 10
*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			



MGU-B.VOC. (HONOURS)

Syllabus

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	3	2	1	0	2	2	1	1	3
CO 2	3	3	3	1	1	2	2	1	2	3
CO 3	3	3	3	1	1	3	3	2	2	3
CO 4	3	3	3	2	2	3	3	2	3	3

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	
	Introduction of Photovoltaic (PV) Technology			
1	1.1	Overview of Photovoltaic (PV) Technology- Introduction, History and Evolution of PV Technology	5	1
	1.2	Basic Principles of Solar Energy Conversion, Types and Modules	6	1
	1.3	PV Materials and Manufacturing Processes	4	1
	1.4	PV System Components and Configurations- Inverters, Charge Controllers, Different kinds of battery technology - Tubular, SMF, Li-ion Battery		1
2	PV System Configurations			
	2.1	PV System Configurations: On-grid, Off-grid, and Hybrid Systems	2	2
	2.2	PV System Design- Site Assessment, Solar Resource Analysis, System Sizing, Design, and Performance Estimation	5	2

2	2.3	MPPT Basic- MPPT Design with PO Algorithm, IC Algorithm, Fuzzy logic (Basic Ideas only)	5	2
	2.4	Electrical Wiring and Connection in Solar Installations. Safety Practices, Regulations, Economic and Environmental Aspects of Solar Power	3	2
Basics of Solar PV Powered				
3	3.1	Basics of Solar PV Powered Electric Vehicle System, Design and components of the solar water pumping system	5	3
	3.2	Performance Monitoring, Data Analysis, Maintenance and Troubleshooting of Solar PV Systems	5	4
	3.3	Emerging Trends and Innovations in Photovoltaics. Case Studies of Successful PV Implementations	5	4
4		Teacher specific content		

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Subject	B.Voc.(Honours) Industrial Instrumentation and Automation			
Course Name	Automation Controllers			
Type of Course	MPC			
Course Code	MG6MPCIIM300			
Course Level	300			
Course Summary	This course provides a comprehensive introduction to Programmable Logic Controllers (PLCs) and Supervisory Control and Data Acquisition (SCADA) systems, which are fundamental technologies in industrial automation and process control. Students will learn the history, architecture, and operation principles of PLCs, including the various hardware components such as CPUs, discrete and analog I/O modules, memory types, and programming devices. The course covers practical aspects like wiring standards, programming techniques, and troubleshooting PLC systems			
Semester	6	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	
		4	0	60
Pre- requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Identify and explain the various parts of a PLC including the CPU, memory units, power supply, and I/O modules.	U	1,2,3, 4,10
2	Apply Boolean algebra rules to simplify logic expressions for digital and PLC applications.	A	1,2,3, 4,10
3	Identify various types of input and output devices used in industrial automation and describe their roles in a PLC-based control system.	U	1,2,3, 4,10
4	Understand the fundamental concepts, components, and architectures of wired and wireless communication systems	U	1,2,3, 4,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2	1	1	0	0	0	0	0	2
CO 2	3	3	1	1	0	0	0	0	0	2
CO 3	2	2	1	1	0	0	0	0	0	2
CO 4	2	2	1	1	0	0	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Syllabus

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	PLC Programmable Logic Controllers (PLCs):			1
	1.1	Definition & history of the PLC; Principles of Operation; Various Parts of a PLC	5	
	1.2	The I/O section, Discrete I/O Modules, Analog I/O Modules, Special I/O Modules, I/O specifications, UAVs for crop monitoring, spraying, and data collection.	5	
	1.3	Types, Programming Devices, Selection of wire types Introduction; The CPU, Memory design, Memory .	5	
2	Fundamentals of Logic			2
	2.1	The Binary Concept, AND, OR and NOT functions, Boolean Algebra, developing circuits from Boolean Expression expressions,	5	
	2.2	Producing the Boolean equation from given circuit, Hardwired logic versus programmed logic, Programming word level logic instructions	5	
	2.3	Converting Relay schematics and Boolean equation into PLC Ladder Programs, Writing a ladder logic program directly from a narrative description	5	

	Various INPUT /OUTPUT Devices and its interfacing with PLC.			
3	3.1	Different types of Input devices: Switches: Push button Switches, Toggle Switches, Proximity switches, Photo switches,	5	3
	3.2	Temperature Switch, Pressure Switch, and Level Switch, Flow Switches, manually operated switches, Motor starters	5	
	3.3	Transducers and sensors, Transmitters Their working, specification and interfacing	5	
	SCADA			
4	4.1	Definition and need for SCADA Evolution of SCADA systems	5	4
	4.2	Functions and features of SCADA.. Applications of SCADA in various industries	5	
	4.3	Architecture	5	
5		Teacher's specific		

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation And Automation				
Course Name	Industrial Safety Measures				
Type of Course	VAC				
Course Code	MG6VACIIM300				
Course Level	300				
Course Summary	This course focuses on the essential principles, practices, and techniques involved in ensuring safety in industrial environments. It covers a wide range of safety measures designed to protect workers, machinery, and the environment from potential hazards, accidents, and incidents.				
Semester	6	Credits		3	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	0	0	60
Pre- requisites, if any	NIL				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Understand the evolution of modern safety concepts,	U	1,2,3,4,5,6,7,8,9,10
2	Learn the concept of an accident and differentiate between reportable and non-reportable accidents	A	1,2,3,4,5,6,7,8,9,10
3	Identify training needs and select appropriate training methods, including programs, seminars, conferences, and safety competitions.	U	1,2,3,4,5,6,7,8,9,10
4	Apply safety management systems and techniques to prevent accidents and promote a safe working environment.	A	1,2,3,4,5,6,7,8,9,10

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	2	2	1	1	2	2	2	1	2
CO 2	3	2	1	1	1	3	2	3	1	2
CO 3	3	3	2	1	1	2	1	2	1	2
CO 4	3	3	2	1	1	2	1	2	1	2

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Evolution of modern safety concept			
	1.1	Safety policy - Safety Organization - line and staff functions for safety	5	1
	1.2	Safety Committee- budgeting for safety. Techniques: Incident Recall Technique (IRT),	5	1
	1.3	Disaster control, Job Safety Analysis (JSA), safety survey, safety inspection, safety sampling, Safety Audit	5	1
2	Accident investigation			
	2.1	Concept of an accident, reportable and non reportable accidents	5	2
	2.2	unsafe act and condition – principles of accident prevention	5	2
	2.3	Supervisory role- Role of safety committee - Accident causation models	5	2
Module	Units	Course description	Hrs	CO No.
3	Safety education and training			
	3.1	Importance of training- identification of training needs- training methods – programme, seminars, conferences, competitions – method of promoting safe practice	5	3
	3.2	role of government agencies and private consulting agencies in safety training – creating	5	4

		awareness, awards, celebrations, safety posters -		
	3.3	safety displays, safety pledge, safety incentive scheme, safety campaign – Domestic Safety and Training.	5	4
4	Teacher's Specific content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) • Use of ICT tools in conjunction with traditional classroom teaching methods • Interactive sessions • Class discussions
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) CCA for Practical(25 Marks) 1. Lab involvement 2. Record 3. Viva
	B. End Semester Evaluation (ESE) ESE for Practical : Practical Examination (50 Marks, 1.5 hrs) Part A : Short Answer Questions(5 out of 7 Questions) - (5*4=20 Marks) Part B : Practical Question (1 Question) – (1*20=20Marks) Record : 10 Marks

REFERENCES:

1. National Safety Council. *Accident Prevention Manual for Industrial Operations*. N.S.C., 1982.
2. Heinrich, H. W. *Industrial Accident Prevention*. McGraw-Hill, 1980.

SUGGESTED READINGS

1. Krishnan N.V., *Safety Management in Industry*, Jaico Publishing House, Bombay, 1997



MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation				
Programme	B.Voc.(Honours) Industrial Instrumentation And Automation				
Course Name	Project				
Type of Course	PRJ				
Course Code	MG6PRJIIM300				
Course Summary	The project work provides students with an opportunity to identify, analyze, and solve real-world problems relevant to their field of study by integrating and applying the theoretical knowledge and skills acquired throughout their academic program. It fosters independent research, critical thinking, innovation, and the practical use of methodologies, tools, and techniques to design effective solutions. Students are encouraged to work individually or in teams, enhancing their collaboration, time management, ethical responsibility, and self-directed learning. The project also develops competencies in academic writing, documentation, and technical communication. Each project is expected to culminate in a comprehensive report, a working model or prototype (where applicable), and a formal presentation followed by a viva voce examination, demonstrating the student's ability to apply knowledge creatively and professionally in a real-world context.				
Semester	6	Duration	8 hours/week	Credits	4

COURSE OUTCOME:

CO No:	Expected Course Outcome	Learning Domains	PO No:
Upon the successful completion of the course, the student will be able to			
1	Identify, analyze, and define problems relevant to the field of study.	An	1,2,3
2	Apply appropriate methodologies, tools, and techniques to design and implement effective solutions.	C	2,3,10
3	Demonstrate skills in research, critical thinking, project planning, and systematic execution.	S	1,2,5,10
4	Produce well-structured academic reports and communicate project outcomes effectively.	S	4,8,10
5	Exhibit teamwork, time management, ethical responsibility, and initiative in a self-directed project environment.	S	5,8,9,10
6	Address real-world challenges with innovative and context-aware solutions.	Ap	1,2,6,10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	2	3	1	0	0	0	0	0	0	0
CO 2	0	3	2	0	0	0	0	0	0	2
CO 3	3	3	0	0	2	0	0	0	0	3
CO 4	0	0	0	2	0	0	0	3	0	3
CO 5	0	0	0	0	3	0	0	2	1	1
CO 6	3	2	0	0	0	1	0	0	0	1

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).

Assessment Types	MODE OF ASSESSMENT		
	A	Continuous Comprehensive Assessment (CCA)	
		Components	Marks
		Commitment and Involvement	5
		Periodic progress review	10
		Quality of work/Implementation effort	10
		Report	5
		Total	30
	B	End Semester Evaluation (ESE)	
		Components	Marks
		Problem Identification and Objectives	10
		Methodology / Design / Technical Content	15
		Implementation / Analysis / Results	15
		Final Report	10
		Presentation	10
		Viva Voce	10
		Total	70

SEMESTER: VII & VIII



MGU-B.VOC. (HONOURS)

Syllabus

Semesters: 7 & 8

B.Voc.Honours

Course Code	Title of the Course	Type of the Course	Credits	Number of Days	Credit Distribution		
					L	P	O
MG7APPIIM400	Apprenticeship	APP	28	280	0	28	0
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On the Job Training

MGU-B.VOC. (HONOURS)

B.Voc.Honours with Research

Course Code	Title of the Course	Type of the Course	Credits	Number of Days	Credit Distribution		
					L	P	O
MG7RINIIM400	Research Internship	RIN	20	200	0	20	0
NA	Online	SDC	4	NA			
NA	Online	SDC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			
NA	Online	MPC	4	NA			

L — Lecture, P — Practical/Practicum , O — On the Job Training



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation
Programme	B.Voc.(Honours) Industrial Instrumentation And Automation
Course Name	Apprenticeship
Type of Course	APP
Course Code	MG7APPIIM400
Course Level	400
Course Summary	As an integral component of the B.Voc. Honours degree programme, students are required to complete a structured apprenticeship or work-integrated learning programme in collaboration with relevant industries, organizations, or institutions. This component, spanning a duration of 280 days, carries 28 academic credits and is compulsory in the student's designated skill domain. It is designed to enhance industry preparedness by reinforcing academic knowledge through sustained, domain-relevant practical experience. The apprenticeship offers students the opportunity to engage directly with real-world professional environments, enabling them to apply domain-specific competencies, gain exposure to industry-standard tools and practices, and participate meaningfully in ongoing operations and projects. This extended, immersive experience serves to bridge the gap between theoretical learning and professional expectations, thereby fostering critical skills for career development and employability. To ensure the effectiveness, academic relevance, and accountability of the apprenticeship: • Each student will be assigned an academic mentor from the parent institution and an industry supervisor from the host organization. • Students are required to maintain a weekly activity logbook, which must be regularly reviewed and signed by the industry supervisor. • Monthly progress reports will be submitted to and reviewed by the academic mentor in consultation with the industry supervisor. • Mid-term and final evaluations will be conducted based on a combination of employer feedback, student outputs/deliverables, and academic performance metrics. The institution will conduct site visits, virtual check-ins, or regular follow-ups to ensure student engagement, address issues promptly, and uphold the quality of

	the apprenticeship experience. This structured apprenticeship is a critical step in preparing students for the dynamic demands of the professional world, ensuring that their academic journey culminates in a well-rounded and industry-aligned skill set.				
Semester	7&8	Duration	280 days	Credits	28

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learnin g Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Gain hands-on professional experience by engaging in long-term, domain-specific apprenticeship in real-world industry environments.	S	1,3,6,10
2	Apply domain-specific theoretical knowledge to solve real-time problems, enhancing technical and problem-solving competencies.	A	1,2,3,10
3	Demonstrate professional competencies such as workplace etiquette, communication skills, and teamwork in a collaborative work culture.	S	4,5,8,9
4	Build a professional portfolio by achieving practical outcomes and establishing credible industry references and credentials.	C	5,9,10
5	Cultivate reflective thinking, adaptability, and a lifelong learning mindset through structured and mentored work experience.	Ap	1,6,8,10
6	Transition smoothly from academic study to professional practice by developing job-specific skills and industry-aligned competencies.	S	2,3,5,10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

	MODE OF ASSESSMENT		
	A	Continuous Comprehensive Assessment (CCA)	
		Components	Marks
		Commitment, Punctuality & Professional Conduct	10
		Monthly Progress Reviews & Logbook Maintenance	25
		Skill Development & Application	25
		Interim Report	20
	Total		90
Assessment Types	B	End Semester Evaluation (ESE)	
		Components	Marks
		Feedback & Evaluation Report from Host Organization	50
		Skill Demonstration / Summary of Work Exposure	40
		Final Report / Learning Portfolio	40
		Domain Knowledge and Experience Communication (Presentation)	40
		Viva Voce	40
		Total	210

Note:

This assessment framework is intended as a guiding structure for evaluating apprenticeship performance. However, in order to remain responsive to the evolving needs of industry and society, the evaluation criteria may be revised from time to time. Such changes aim to enhance the relevance, effectiveness, and fairness of the assessment process.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation			
Programme	B.Voc.(Honours) Industrial Instrumentation And Automation			
Course Name	Research Internship			
Type of Course	RIN			
Course Code	MG7RINIIM400			
Course Level	400			
Couse Summary	As an integral requirement of the B.Voc. Honours with Research degree programme, the Research Internship is designed to provide students with hands-on exposure to real-world research practices in their designated skill domain. This component carries 20 academic credits and extends over a duration of 200 days. The internship must be undertaken in collaboration with a research organization, industry, or university department, under the mentorship of a qualified research guide. The primary aim of this internship is to engage students in industry-linked research projects that allow them to apply theoretical knowledge to practical, domain-specific problems. Students are expected to work on meaningful research inquiries, contribute to data collection and analysis, develop critical thinking and problem-solving skills, and enhance their communication and documentation abilities. In addition to the research internship, students must earn 8 credits through Skill Development Courses (SDCs), specifically chosen for their research orientation, thereby reinforcing their academic and practical foundation. This component not only contributes significantly to the academic rigor of the Honours with Research degree but also ensures a seamless transition from classroom learning to workplace research, preparing students for advanced studies or professional roles in their respective domains.			
Semester	7&8	Duration	200 days	Credits
				20

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Demonstrate research aptitude and inquiry-based learning by actively engaging in real-time research projects.	S	1,2,10
2	Apply academic knowledge in a professional research environment to bridge the gap between theory and real-world research practices.	A	2,3,6,10
3	Strengthen domain-specific knowledge and technical competencies through systematic investigation and practical application.	S	1,2,3
4	Address real-world research problems using problem-solving, analytical, and critical thinking skills.	S	1,2,6
5	Communicate scientific ideas and findings effectively through research reports, documentation, and presentations.	S	4,8,10
6	Collaborate with researchers and peer groups to gain exposure to interdisciplinary perspectives and collaborative learning practices.	S	2,3,5,10
7	Demonstrate professional growth and readiness for higher education, entrepreneurship, or research-oriented careers.	I	5,9,10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

Assessment Types	MODE OF ASSESSMENT		
	A	Continuous Comprehensive Assessment (CCA)	
		Components	Marks
		Commitment, Punctuality & Professional Conduct	10
		Monthly Progress Reviews & Logbook Maintenance	15
		Skill Development & Application	15
		Interim Report	20
		Total	60
	B	End Semester Evaluation (ESE)	
		Components	Marks
		Feedback & Evaluation Report from Host Organization	40
		Skill Demonstration / Summary of Work Exposure	20
		Final Report / Learning Portfolio	25
		Domain Knowledge and Experience Communication (Presentation)	25
		Viva Voce	30
		Total	140

Note:

This assessment framework serves as a guiding structure for evaluating research internship performance. However, to remain responsive to the evolving needs of industry, academia, and society, the evaluation criteria may be revised periodically. Such revisions aim to enhance the relevance, effectiveness, and fairness of the overall assessment process.



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Technology and Applied Sciences / Industrial Instrumentation and Automation
Programme	B Voc.(Honours) Industrial Instrumentation And Automation
Course Name	On-the-Job Training
Type of Course	SDC
Course Summary	On-the-Job Training program aims to provide undergraduate students specializing in Industrial Instrumentation with hands-on experience in professional environments such as manufacturing plants, process industries, calibration laboratories, and automation sectors. Through this program, students will strengthen their competencies in instrumentation techniques, sensor calibration, process control, data acquisition, system troubleshooting, and technical documentation. They will gain practical knowledge of industrial practices, exposure to modern instrumentation tools and standards, and develop valuable industry connections. The program also seeks to foster collaboration between academic institutions and industries/research organizations, enabling students to apply theoretical knowledge to real-world industrial scenarios.

COURSE OUTCOMES (CO)

CO No:	Expected Course Outcome	Learning Domains	PO No:
	Upon the successful completion of the course, the student will be able to		
1	Demonstrate understanding of industry operations, standards, and professional expectations through direct exposure to workplace environments.	Ap	1,3,6,10
2	Apply job-specific skills effectively in real-world tasks and responsibilities within the assigned industry setting.	S	2,4,5,10
3	Integrate academic knowledge with practical applications to solve work-related challenges and contribute to organizational goals.	An	1,2,3,6
4	Exhibit essential workplace competencies such as punctuality, accountability, communication, teamwork, and adaptability.	S	4,5,8,9

5	Identify and evaluate potential career opportunities by reflecting on their internship experiences and professional interactions.	E	1,9,10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)			

	MODE OF ASSESSMENT		
	A	Continuous Comprehensive Assessment (CCA)	
		Components	Marks
		Feedback from the hosting organisation	5
		Internal supervisor feedback	10
		Total	15
	B	End Semester Evaluation (ESE)	
		Components	Marks
		Presentation	10
		Report	10
		Viva Voce	15
		Total	35
Assessment Types			

MGU-B.VOC. (HONOURS)

Syllabus